

Form 61D

(June 2005)

ACT Assessment

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In response to your recent request for test information release materials, this booklet contains the test questions and conversion tables used in determining your ACT scores. Enclosed with this booklet is a report listing your answers to the ACT Assessment tests and the answer key.

If you wish to order a photocopy of your answer document—including, if you took the Writing Test, a copy of your written essay—please use the order form on the inside back cover of this booklet.

We hope that you will find this information helpful.

ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for the underlined part. In most cases, you are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose "NO CHANGE." In some cases, you will find in the right-hand column a question about the underlined part. You are to choose the best answer to the question.

You will also find questions about a section of the passage, or about the passage as a whole. These questions do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. For many of the questions, you must read several sentences beyond the question to determine the answer. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

Finding Tomatoes in Winter

[1] When I was growing up on our farm in North Carolina, my favorite chore was canning tomatoes with my mother. [2] In our one-acre garden, the weight of the ¹ tomato plants would slump with ripening fruit, the stems would break, and the pungent smell of late summer would be released. [3] Pleasantly heavy and warm ² from the sun, I can still imagine the feel of the fruit. [4] That smell told us that it was time to walk the vines and the best of the crop being selected for canning. ³ 4

1. A. NO CHANGE
B. size of the
C. ripening
D. OMIT the underlined portion.
2. F. NO CHANGE
G. I can still imagine the feel of the fruit, pleasantly heavy and warm from the sun.
H. Pleasantly, the fruit is still warm and heavy where I can feel it in my imagination from the sun.
J. Pleasantly heavy, I can still imagine the feel of the warm fruit from the sun.
3. A. NO CHANGE
B. we select the best of the crop for canning.
C. select the best of the crop for canning.
D. canning the best of the crop that we selected.
4. For the sake of the logic and coherence of this paragraph, Sentence 4 should be placed:
F. where it is now.
G. before Sentence 1.
H. after Sentence 1.
J. after Sentence 2.

We washed the tomatoes in the kitchen while a large pot of water heated up on the stove. ⁵ When the water came to a boil, we dropped the tomatoes into the pot for exactly one minute, retrieved them, and then plunged them into ice water. This double dunking loosened the skin so we could easily peel it away from the juicy, red pulp.

Next, my mother and I, cored the tomatoes. Then ⁶ came my favorite part: crushing them into small pieces to fit into the canning jars. I can see myself now, standing in a steamy kitchen, overalls spattered with pulp, sweat ⁷ dripping from my chin. It was impossible to keep a straight face as seeds squirted through the air and stuck to the walls around us. Sometimes ⁸ we sang along with the radio, frequently substituting

the word *tomato* for a prominent word in the song. ⁹ Sometimes the only sound in the kitchen would be the fan,

rotating on its' squeaky ¹⁰ hinge, sending a breeze our way.

Patches of salt forming ¹¹ on our faces and arms as our sweat evaporated into the August air.

5. If the writer were to delete the phrase "while a large pot of water heated up on the stove" from the preceding sentence (placing a period after the word *kitchen*), the paragraph would primarily lose a phrase that:
 - A. describes the duties specific to the narrator's role in canning tomatoes.
 - B. indicates where the canning takes place and why water is needed in the process.
 - C. establishes a logical transition to the information in the remainder of the paragraph.
 - D. establishes that part of the canning process involved putting the tomatoes in water.
6. F. NO CHANGE
 G. Next my mother and I,
 H. Next my mother, and I
 J. Next, my mother and I
7. A. NO CHANGE
 B. overalls spattered, with pulp sweat;
 C. overalls, spattered with pulp, sweat
 D. overalls spattered with pulp sweat
8. Given that all the choices are true, which one helps the reader best visualize the specifics of the scene?
 - F. NO CHANGE
 - G. air while we watched with amusement.
 - H. air, creating a sight to behold.
 - J. air on those hot summer days.
9. The writer is considering deleting the preceding sentence. Should the writer make this change?
 - A. Yes, because the sentence disrupts the paragraph's attention to the visual aspects of the narrator's memory of canning tomatoes.
 - B. Yes, because the sentence inappropriately shifts the focus of the paragraph to the narrator's enthusiasm for singing.
 - C. No, because the sentence is consistent with the paragraph's focus on the aspects of canning tomatoes that are memorable to the narrator.
 - D. No, because the sentence sets up a logical contrast with the narrator's reserved behavior as described in the sentence before it.
10. F. NO CHANGE
 G. it's squeakier
 H. its squeaky
 J. it's squeaky
11. A. NO CHANGE
 B. would form
 C. form
 D. OMIT the underlined portion.

Once the chore was finished, the tomatoes would line the pantry shelves alongside spiced peaches and snap beans, all gleaming colorfully in their jars. Together they

12

were our familys' protection against the bleak winter.

13

Now I live a long way from my family and their farm in North Carolina. In the winter, I savor the fresh supply of the half dozen or so jars of tomatoes Mom sends me every year. On special occasions, I reach for one, twist off the top, and let the smell of summer drift back to me

14

from my childhood. [15]

12. Given that all the choices are true, which one establishes a visual contrast with the image in the final sentence of this paragraph?

F. NO CHANGE
G. each jar the same size as the next.
H. giving me a sense of real accomplishment.
J. sometimes as many as twenty jars of each.

13. A. NO CHANGE

B. family's
C. families
D. families'

14. F. NO CHANGE

G. with
H. before
J. along

15. If the writer were to delete this final paragraph, the essay would primarily lose:

A. images that reveal that the narrator's memories of canning tomatoes are fading over time.
B. a sense that the narrator is planning to move back to North Carolina as soon as possible.
C. information that indicates that distance contributes to the narrator's nostalgia about canning tomatoes.
D. information about the narrator's adult life that is irrelevant to the essay's focus on canning tomatoes.

PASSAGE II

A Tree's Change in Status

[1]

[1] For almost two millennia, the yew an evergreen tree with a poisonous fruit and a twisted trunk, was scorned or feared by many people, particularly Europeans. [2] In the first century A.D., the Greek biographer Plutarch warned of sleeping in the shade of a yew was a deadly practice. [3] About the same time, the Roman author Pliny pronounced the yew "unpleasant and fearful to look upon." [4] The old

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16. F. NO CHANGE

G. yew, an evergreen tree, with a poisonous fruit
H. yew, an evergreen tree with a poisonous fruit
J. yew an evergreen tree with a poisonous fruit,

17. A. NO CHANGE

B. warned people about
C. having warned that
D. warned that

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

British practice of planting yews in cemeteries did nothing in changing or altering that reputation.

[5] Recently, however, the discovery of taxol, a substance found in the bark of the yew, has given the tree new status. [19]

[2]

In the 1960s, scientists discovered that it was a promising anticancer drug. [21] Although pharmaceutical

companies that begun to produce the drug learned that taxol could be extracted from the bark of all species of yew, its attention was soon focused on

the Pacific yew. [24]

[3]

To produce enough taxol to provide for the needs of all patients, chemists estimated that they would need the

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18. F. NO CHANGE
G. to alter that acquired
H. in the way of changing that
J. to alter that

19. For the sake of the logic and coherence of this paragraph, Sentence 2 should be placed:

A. where it is now.
B. before Sentence 1.
C. after Sentence 3.
D. after Sentence 4.

20. F. NO CHANGE
G. that taxol was
H. it as
J. this was

21. At this point, the writer is considering adding the following true statement:

There were many medical breakthroughs during the 1960s, and this was only one of them.

Would this be a relevant addition to make here?

- A. Yes, because it provides an important link to the rest of the paragraph.
B. Yes, because it informs the reader that taxol is not the only such drug available.
C. No, because it strays from the main focus of the essay.
D. No, because it implies that scientists did not understand why taxol was important.

22. F. NO CHANGE
G. who'd begun producing
H. who begun producing
J. that began to produce

23. A. NO CHANGE
B. it soon focused its attention
C. they soon focused their attention
D. it was soon focused

24. Given that all of the following are true, which one, if added here, would most effectively conclude Paragraph 2 and provide a link to Paragraph 3?

F. Natural stands in the Pacific Northwest represented the largest available source of yew bark.
G. In western North America, the yew grows in cool, moist climates.
H. The Pacific yew is very shade-tolerant and prefers upland sites or riversides.
J. The yew tree, *Taxus brevifolia*, will grow to fifty feet in height and reach a diameter of two feet.

25. A. NO CHANGE
B. Enough taxol production to provide
C. Producing enough taxol in providing
D. They wanted to produce enough taxol to provide

bark from about 40,000 Pacific yews each year. Since the
²⁶
 yew is slow growing and sparsely distributed among other

trees in the forest, it is scattered throughout the area. To
²⁷

save the trees and still provide for the patient's needs,
²⁸
 scientists began to produce taxol synthetically in the
 laboratory from the tree's leaves. This process is more
²⁹
 costly than harvesting the bark, growing the foliage in a
 nursery provides a renewable resource.

[4]

Recent studies have shown that Pacific yews can be
 successfully propagated at a minimal cost. Whether the
³⁰
 young plants are kept under conditions with high humidity,
 their survival rate is about 70 percent. These studies
 indicate that chemists will be able to secure an adequate
 supply of taxol and also save the Pacific yew from
 possible extinction.

[5]

With the discovery of taxol, the tree that was once
 associated with death has now become a tree of life.

PASSAGE III

The \$275,000 Sack of Flour

In April 1864, during the Civil War, the
 Nevada mining town of Austin held its first mayoral
 election. Mr. Reuel Gridley, a grocer, entered into a

26. Which of the following alternatives to the underlined portion would NOT be acceptable?
- F. of about
 - G. for approximately
 - H. of around
 - J. from approximately
27. Given that all of the choices are true, which one would provide an essential link to the sentence that follows?
- A. NO CHANGE
 - B. its reproduction is through birds, which scatter the seeds in many different places.
 - C. American conservationists began to fear that the species would be irreparably depleted.
 - D. this does not diminish the fact that the wood is tough and has other uses.
28. F. NO CHANGE
 G. needs' of the patients,
 H. patients' needs,
 J. patients needs,
29. A. NO CHANGE
 B. This
 C. As this process
 D. Although this process
30. F. NO CHANGE
 G. That
 H. Seeing
 J. If

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

friendly wager for the election's outcome with a
31

Dr. Buel Herrick. Nevertheless, that wager would
32

benefit many Civil War soldiers in an unforeseen way.
33

The bets terms stated that the loser of the
34

wager, escorted by a marching band, must carry
35
a fifty-pound sack of flour one mile along Austin's Main

Street. While voters elected the candidate that Dr. Herrick
36
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supported, choosing that person to be mayor.
37

Mr. Gridley, the loser, then decorated a sack of flour
with red, white, and blue ribbons: shouldered it; and led
38
the band through Austin. At the parade's end, he offered
the flour to the highest bidder, starting with his own \$200
bid. He stated that proceeds would go to the U.S. Sanitary
Commission, an organization equivalent and similar to
39
today's Red Cross, to aid Civil War soldiers.

Bidding became brisk. Townspeople felt compassion
40
for the soldiers fighting in the eastern United States.

31. A. NO CHANGE
B. of
C. on
D. at

32. F. NO CHANGE
G. Moreover,
H. Consequently,
J. Eventually,

33. Which of the following alternatives to the underlined
portion would be LEAST acceptable?
A. assist
B. enhance
C. serve
D. help

34. F. NO CHANGE
G. bets terms'
H. bet's terms
J. bets' terms

35. A. NO CHANGE
B. wager escorted by a marching band
C. wager escorted by a marching band,
D. wager, escorted by a marching band

36. F. NO CHANGE
G. Since voters
H. When voters
J. Voters

37. A. NO CHANGE
B. supported and picked that person to become
mayor.
C. supported as their choice for mayor of the town.
D. supported.

38. F. NO CHANGE
G. ribbons, shouldered it,
H. ribbons shouldered it,
J. ribbons, shouldered it;

39. A. NO CHANGE
B. the equivalent of
C. a similar equivalent of
D. the self-same organization equivalent to

40. Which of the following alternatives to the underlined
portion would NOT be acceptable?
F. brisk, townspeople
G. brisk because townspeople
H. brisk, since townspeople
J. brisk; townspeople

The successful bidder donates the sack of flour back to

41

Gridley to be auctioned off again. Each time that Gridley
started the bidding anew. In two days, the sack sold
112 times, for amounts ranging from \$10 to \$350.

Gridley collected \$4,350 for the Sanitary Commission.
By comparison, Sanitary Commission donation boxes
left at voting polls had received only \$200.

Being inspiring and motivating himself, Gridley
traveled to Virginia City, Nevada, to auction the sack

43

of flour to its residents. Money poured in. Gridley
trekked to California: San Franciscans donated \$2,800,
and Sacramento's citizens gave \$10,000. Gridley then
proceeded to St. Louis and to major eastern cities. These
bidders added \$170,000 to the Sanitary Commission's
fund. Within twelve months Gridley's sack of flour had
earned \$275,000.

45

Reuel Gridley died penniless in 1870. In 1914 family
members donated the famous flour sack to the Nevada
Historical Society in Reno, where it remains, a reminder
of his hard work and the generosity of many.

41. A. NO CHANGE
B. could of donated
C. can donate
D. donated

42. F. NO CHANGE
G. time, Gridley
H. time that he
J. time when he

43. A. NO CHANGE
B. Himself being inspired and motivated,
C. By inspiring and motivating,
D. Inspired and motivated,

44. F. NO CHANGE
G. his
H. their
J. there

45. A. NO CHANGE
B. fund. Likewise, within
C. fund, because within
D. fund, while within

PASSAGE IV

The Nguyen Family Restaurant

For my friends Trần and Maya Nguyen,
the distance between Vietnam and California

46

is more than it's measure in miles.

47

46. F. NO CHANGE
G. among
H. from
J. to

47. A. NO CHANGE
B. its'
C. its
D. his

Although immigrants, Tran, Maya, and

⁴⁸ their four children left much behind, including

their family restaurant. Now they want to run a
⁴⁹ restaurant in Los Angeles just as they did in Da Nang.

When Tran visited Vietnamese restaurants in the
Los Angeles area, he was impressed, with both, their
⁵⁰ decor and their menus but shocked to learn how

much it costs to rent a suitable building. 51 He

knew that himself and Maya would have to
⁵²

apply their energies to the activity of labor
⁵³ for the money needed to sign a lease.

During their first year in the United States, they
found jobs and began to save money. Tran worked as
a hospital clerk, and Maya was an assembler in an
⁵⁴

electronics factory. Nevertheless, they were a long way
⁵⁵

48. F. NO CHANGE

- G. Concerning immigration,
- H. After immigrating,
- J. As immigrants,

49. A. NO CHANGE

- B. In conclusion,
- C. Likewise,
- D. Instead

50. F. NO CHANGE

- G. impressed with both
- H. impressed with both,
- J. impressed, with both

51. The writer is considering deleting the phrase "to rent a suitable building" from the preceding sentence. If the writer were to make this deletion, the paragraph would primarily lose:

- A. details that help indicate the actual and specific rental cost of a building.
- B. information essential to understanding exactly what the phrase "it costs" refers to.
- C. information that explains the relationship between the decor and the rent of a building.
- D. nothing of significance, since that information is presented earlier in the paragraph.

52. F. NO CHANGE

- G. Maya and himself
- H. Maya and him
- J. he and Maya

53. A. NO CHANGE

- B. employ themselves vigorously to the task at hand
- C. work a long time
- D. accomplish it

54. Which of the following alternatives to the underlined portion would NOT be acceptable?

- F. clerk, while
- G. clerk where
- H. clerk;
- J. clerk.

55. A. NO CHANGE

- B. Furthermore,
- C. Eventually,
- D. Besides,

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

from what they would need. They knew their jobs would
help them to raise the necessary funds. After two years,
they had more than doubled their savings but were still
short of their goal.

Tran worked hard at his job. He decided to rent a food
vendor's cart that he could manage on his days off from
the hospital. He knew the perfect location on a busy street
corner. In less than a week, Tran was selling *cha gio* (egg
rolls) on that corner. Soon they rented a second cart for
Maya to operate. This undertaking wasn't their dream but
perhaps their dream would lead to it.

Recently, Maya saw what seemed to her the perfect
building for rent. The neighborhood was friendly,
with lots of foot traffic. She envisioned their restaurant
fitting in nicely here. But she was disappointed

when she talked to the real estate agent—the building
wouldn't be available for six months. Maya went home
and discussed the situation with Tran. They were
discouraged but realized that they could use this time
to acquire much of the equipment they would need.

Last night they decided to name their future restaurant
after their son, Haing. That was also the name of their
restaurant back in Vietnam.

56. At this point, the writer wants to provide a specific
example of the efforts being made by the Nguyens to
reach their goal. Given that all the choices are true,
which one would best accomplish this purpose?

- F. NO CHANGE
- G. Meanwhile, the children adapted to and worked
hard in their new schools.
- H. They postponed buying a car, relying on city buses
for transportation.
- J. Tran and Maya realized that these jobs were means
to a worthwhile end.

57. Which choice would most effectively introduce this
new paragraph?

- A. NO CHANGE
- B. Their goal meanwhile seemed just as unattainable.
- C. Maya encouraged Tran not to lose hope.
- D. Then one day Tran had an inspiration.

58. F. NO CHANGE
G. they would lead to that.
H. would lead it to that.
J. would lead to it.

59. A. NO CHANGE
B. traffic, that she
C. traffic, she
D. traffic she

60. Which of the following alternatives to the underlined
portion would NOT be acceptable?

- F. agent. The
- G. agent: the
- H. agent, the
- J. agent; the

PASSAGE V

A Night at the Vanguard

My friend Stacia and I recently took a class in jazz music, so one night we went down to the Village Vanguard to hear a performance, by saxophonist, Jackie McLean.

The Vanguard is a New York City nightclub located underground in a room crowded with tables. ⁶¹ Photos of Billie Holiday, Thelonious Monk, John Coltrane, and

other jazz greats on the walls.

To our surprise, Stacia and I found empty seats near the stage. The backup band of a pianist, a bass player, and the talented Billy Higgins started with a piece ⁶⁴ Stacia and I immediately recognized—"Seven Minds."

My attention focused first with Higgins, who played ⁶⁵ with his head tipped back, smiling and nodding.

Soon McLean stepped onstage with his sax I wasn't ⁶⁶

prepared for the impact. Into every corner of the ⁶⁷

61. A. NO CHANGE
B. performance by saxophonist,
C. performance, by saxophonist
D. performance by saxophonist
62. The writer is considering deleting the preceding sentence. Should the writer make this deletion?
F. Yes, because the sentence detracts from the essay's focus on music.
G. Yes, because the sentence sets up the expectation that the essay will be about New York City nightclubs.
H. No, because the sentence is informative and contributes to the essay's description of the narrator's experience.
J. No, because the sentence suggests that the narrator was unable to enjoy listening to jazz in a club where the seating was uncomfortable.
63. A. NO CHANGE
B. line
C. along
D. hanging on
64. Given that all of the choices are true, which one most clearly adds new information about the mixture of sounds in the music being played the night the narrator visited the Village Vanguard?
F. NO CHANGE
G. famous drummer
H. musician
J. jazz artist
65. A. NO CHANGE
B. initially with
C. first on
D. initially to
66. F. NO CHANGE
G. sax.
H. sax that
J. sax, which
67. A. NO CHANGE
B. ready for the unexpected impact.
C. prepared, regarding the impact, for what happened.
D. ~~expecting~~ the impact, which I didn't see coming.

room, he blew wild, sweet, almost disjointed notes

that seemed to leave many listeners amazed. He

didn't play according to the rules of the popular

music I had grown up listening to on the radio. McLean

has been likened to the sax genius Charlie Parker,

and that night I could hear why.

[1] Occasionally McLean would stroll offstage.

[2] Then one of the other musicians taking the lead, pulling

strange patterns from his instrument; the equivalent in
sound of someone dancing under a blinking strobe light.

[3] Once during a drum solo, a man near us started
laughing, as if Higgins had just told a mischievous joke.

[4] Now and then, one member of the audience or another

would bend forward as if you were intently following
an argument that is both complicated and fascinating.

[5] When he played, something happy happened inside

me. [6] I closed my eyes and just listened. [73]

Suddenly the show was over. The musicians walked
offstage. The room filled with applause. Moments later,
we were outside where the whole city—cabs, couples,
neon lights—seemed to be moving to that bright, hectic,
spontaneous music.

68. At this point, the writer wants to provide an image to help give a sense of the sound of the music being played by McLean. Which choice best accomplishes the writer's purpose?

- F. NO CHANGE
- G. that words cannot hardly describe.
- H. before we knew what was happening to us as lucky members of the audience that night.
- J. that seemed snatched from a dozen different places at once.

69. A. NO CHANGE
B. that was listened to growing up
C. I listened to growing up
D. I listen to when I grew up

70. F. NO CHANGE
G. took the lead, pulling
H. took the lead, he pulled
J. taking the lead by pulling

71. The writer would like to describe the sound of the music in visual terms to help the reader imagine the jazz played that night. Which choice best accomplishes the writer's purpose?

- A. NO CHANGE
- B. the highlight of the evening for some audience members.
- C. the sound making me see why so many people like jazz.
- D. the notes having a visible effect.

72. F. NO CHANGE
G. when it was
H. in the posture that is
J. in the position of someone

73. Upon reviewing this paragraph and realizing that a detail has been left out, the writer composes the following sentence incorporating that detail:

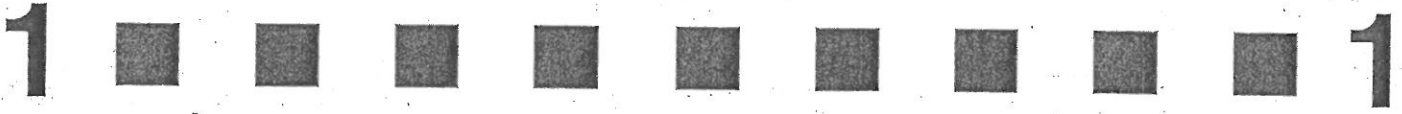
The most intense moment for me came when McLean let loose.

For the sake of the logic and coherence of this paragraph, this sentence should be added after Sentence:

- A. 3.
- B. 4.
- C. 5.
- D. 6.

74. Which of the following alternatives to the underlined portion would NOT be acceptable?

- F. offstage, and the
- G. offstage; the
- H. offstage. Then the
- J. offstage as if the



Question 75 asks about the preceding passage as a whole.

75. The writer is considering adding the following sentence to the essay, placing it before the essay's first sentence as an introduction:

Someday I hope to introduce my family to jazz music.

Should the writer make this addition?

- A. Yes, because the sentence establishes the narrator's enthusiasm for jazz.
- B. Yes, because the sentence clarifies what the essay is about.
- C. No, because the sentence does not add anything significant to the essay.
- D. No, because the sentence contradicts information contained in the essay.

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.



MATHEMATICS TEST

60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

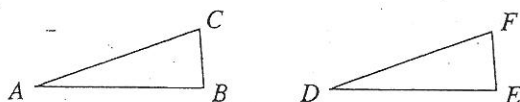
Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. A fingernail grows at an average rate of 0.12 inches per month. Rip Van Winkle, a character in a story by Washington Irving, slept for 20 years. If his fingernails grew at the average rate, approximately how many inches did each grow while he slept?

- A. 28.8
- B. 24.0
- C. 20.0
- D. 12.0
- E. 2.4

2. In the figure below, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\triangle ABC \cong \triangle DEF$. The measure of $\angle A$ is 20° . The measure of $\angle F$ is 75° . What is the measure of $\angle B$?



- F. 20°
- G. 75°
- H. 85°
- J. 95°
- K. 105°

3. Shanika has a bag containing 50 pieces of candy: 6 strawberry, 12 orange, 10 lime, and 22 lemon. If Shanika randomly takes a piece of candy from the bag, what is the probability that the piece she takes is strawberry or lime?

- A. $\frac{6}{50}$
- B. $\frac{10}{50}$
- C. $\frac{16}{50}$
- D. $\frac{16}{34}$
- E. $\frac{60}{50}$

DO YOUR FIGURING HERE.



DO YOUR FIGURING HERE.

4. It is estimated that each 4-ounce portion of a certain type of apple contains 60 calories. What is the estimate for the number of calories contained in a 15-ounce portion of this type of apple?
- F. 15
G. 16
H. 60
J. 225
K. 240
5. Dan and Kyla dined at a restaurant, and their bill was \$29.74. They would like to leave a tip of approximately 15% of their bill. Which of the following is closest to this amount?
- A. \$1.50
B. \$3.00
C. \$3.78
D. \$4.00
E. \$4.50
6. Buyer's Buddy Supermarket is running a sale on all 4 flavors of 2-liter Zing-N-Zest soft drinks. The sale price for each 2-liter bottle is \$0.59, with a limit of 6 bottles per flavor. At most, how many 2-liter bottles can be purchased for \$12.89, excluding tax?
- F. 6
G. 10
H. 21
J. 22
K. 24
7. According to the U.S. Energy Information Administration, in 1989 the U.S. population consumed a total of 81,342 trillion British thermal units (Btu) of energy, including 5,677 trillion Btu of nuclear electric power and 2,884 trillion Btu of hydroelectric power. In scientific notation, how many more Btu of nuclear electric power were consumed than Btu of hydroelectric power?
- (Note: 1 trillion = 1,000,000,000,000)
- A. 2.457×10^{15}
B. 2.793×10^{15}
C. 5.250×10^{15}
D. 7.567×10^{15}
E. 7.846×10^{15}
8. When $3x - 12y = 6$ and $y = -2$, what is the value of x ?
- F. -18
G. -6
H. $\frac{4}{3}$
J. 4
K. 10



DO YOUR FIGURING HERE.

9. Diastolic blood pressure tends to increase linearly with age. Suppose that the average diastolic blood pressure of 20-year-olds is 120 and the average diastolic blood pressure of 60-year-olds is 140. Given this model, what would be the average diastolic blood pressure of 50-year-olds?

A. 120
B. 125
C. 130
D. 135
E. 140

10. The expression $(2x + 3)(3x - 2)$ is equivalent to:

F. $5x + 1$
G. $5x - 1$
H. $6x^2 - 6$
J. $6x^2 - 5x - 6$
K. $6x^2 + 5x - 6$

11. Jan returned a library book that was 9 days overdue. The library charged Jan s cents per day for the first 6 overdue days and 20 cents per day for each overdue day after 6 overdue days. Which of the following is an expression for the total amount, in cents, Jan was charged?

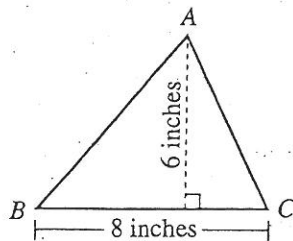
A. $s + 180$
B. $6s + 20$
C. $6s + 60$
D. $9s + 20$
E. $9s + 60$

12. What is the value of $h(1.5)$ when

$$h(x) = -4x^2 + 20x + 32?$$

F. 26
G. 50
H. 53
J. 56
K. 71

13. What is the area, in square inches, of $\triangle ABC$ below?



A. 10
B. 12
C. 14
D. 24
E. 48

14. For what value of x is $3(x + 5) = 2(x - 6)$ true?

F. 27
G. 11
H. -11
J. -21
K. -27



DO YOUR FIGURING HERE.

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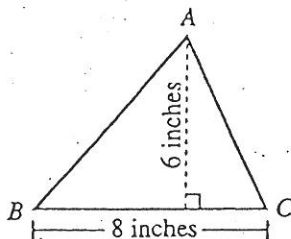
A. $s + 180$
B. $6s + 20$
C. $6s + 60$
D. $9s + 20$
E. $9s + 60$

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E. 48

14. For what value of x is $3(x + 5) = 2(x - 6)$ true?

F. 27
G. 11
H. -11
J. -21
K. -27



15. Which of the following is equivalent to $\frac{25a}{4} + \frac{3a}{2}$?

A. $4\frac{2}{3}a$
 B. $7a$
 C. $7\frac{1}{3}a$
 D. $7\frac{3}{4}a$
 E. $15\frac{1}{2}a$

DO YOUR FIGURING HERE.

16. Suppose that w , y , and z each stand for a digit from 0 through 9 and

$$\begin{array}{r} 5w7 \\ + 92y \\ \hline 1,z96 \end{array}$$

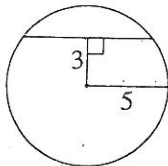
What is the value of the product $w \cdot y \cdot z$?

F. 19
 G. 28
 H. 216
 J. 252
 K. 270

17. In $\triangle ABC$, $\angle B$ is a right angle and the measure of $\angle A$ is 36° . A second right triangle, $\triangle DEF$, is being constructed so that $\angle E$ is a right angle and the measure of $\angle D$ is double the measure of $\angle A$. What is the measure of $\angle F$?

A. 72°
 B. 54°
 C. 30°
 D. 27°
 E. 18°

18. The figure below shows a circle with a radius of 5 cm and a chord of the circle. The chord is 3 cm from the center of the circle. What is the length, in centimeters, of the chord?



F. 3
 G. 4
 H. 5
 J. 8
 K. 10

19. What is the sum of the 2 solutions of the equation $x^2 + 3x - 28 = 0$?

A. -28
 B. -7
 C. -3
 D. 0
 E. 4



20. The time, t seconds, required for a simple pendulum x feet long to make 1 complete swing can be modeled by $t = 2\pi\sqrt{\frac{x}{32}}$. How many seconds are required for 1 complete swing of a simple pendulum that is 8 feet long?

F. Less than 1
 G. Between 1 and 2
 H. Between 2 and 3
 J. Between 3 and 4
 K. More than 4

DO YOUR FIGURING HERE.

21. What is the value of y in the solution to the following system of equations?

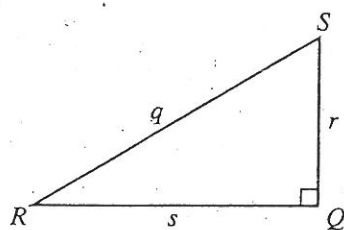
$$\begin{aligned} 2x + y + 3 &= 30 \\ x + 2y &= 30 \end{aligned}$$

A. 1
 B. 8
 C. 11
 D. 17.4
 E. 29

www.crackab.com

22. In the right triangle below, the side lengths given are in inches. Which of the following expressions equals the sine of $\angle R$?

F. $\frac{q}{r}$
 G. $\frac{r}{q}$
 H. $\frac{r}{s}$
 J. $\frac{s}{q}$
 K. $\frac{s}{r}$



23. A regular pentagon, P_1 , has a perimeter of 25 inches, while another regular pentagon, P_2 , has a perimeter that is 15 inches greater than the perimeter of P_1 . The length of each side of P_2 exceeds the length of each side of P_1 by how many inches?

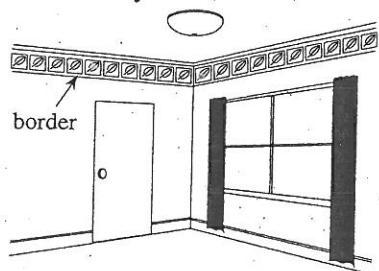
A. 3
 B. 5
 C. 8
 D. 10
 E. 15

www.crackab.com



DO YOUR FIGURING HERE.

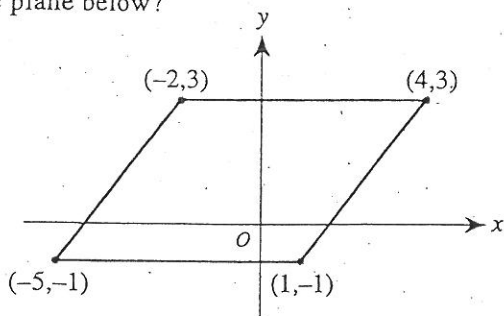
24. A certain type of combination lock has 3 dials: the first 2 dials each have settings for all the digits 0 through 9, and the third has settings for all the 26 capital letters of the alphabet. A combination consists of one setting from each of the dials. (For example, 6-4-P is a combination.) How many different combinations are possible?
- F. 260
G. 520
H. 1,664
J. 2,600
K. 5,200
25. Through how many degrees does a wheel rotate in 1.5 minutes at 45 revolutions per minute?
- A. 24,300
B. 16,200
C. 12,150
D. 8,100
E. 4,050
26. What is the fourth term in the geometric sequence 36, -12, 4, ... ?
- F. $-\frac{4}{3}$
G. -1
H. $-\frac{3}{4}$
J. $\frac{3}{4}$
K. $\frac{4}{3}$
27. $[(a^3)^2(a^2)]^3$ is equivalent to:
- A. a^{10}
B. a^{11}
C. a^{21}
D. a^{24}
E. a^{36}
28. Joe wants to put a border all the way around his bedroom walls 1 inch below the ceiling, as shown below. The border is sold only in whole rolls, and each roll is 5 yards long. Joe's bedroom is rectangular, 12 feet wide and 14 feet long. What is the minimum number of rolls of border Joe must buy?



- F. 2
G. 3
H. 4
J. 5
K. 6



29. What is the area, in square coordinate units, of the parallelogram with vertices as shown in the standard (x,y) coordinate plane below?



- A. 16
B. 18
C. 24
D. 30
E. 36

DO YOUR FIGURING HERE.

30. Suppose that the measure of the larger of 2 supplementary angles is 5 times the measure of the smaller angle. What is the measure of the larger angle?

(Note: In the figure below, $\angle ABD$ and $\angle DBC$ are supplementary. The sum of their measures is 180° .)

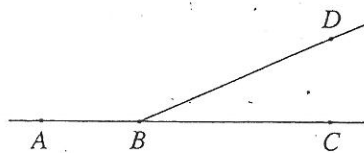
F. 30°

G. 36°

H. $87\frac{1}{2}^\circ$

J. $92\frac{1}{2}^\circ$

K. 150°



31. In the standard (x,y) coordinate plane, what is the slope of the line with the equation $3x - 2y = 24$?

A. -3

B. 3

C. $-\frac{2}{3}$

D. $-\frac{3}{2}$

E. $\frac{3}{2}$

32. The community orchestra will give 18 performances this season. A season ticket costs \$60, and single-performance tickets cost \$5 each. Huong purchased a season ticket. What is the minimum number of performances Huong must attend to save money by buying the season ticket instead of buying single-performance tickets for the performances she attends?

F. 3

G. 4

H. 11

J. 12

K. 13



33. A line in the standard (x,y) coordinate plane passes through the origin and the point $(-12,6)$. An equation of this line is:

A. $y = -x - 6$

B. $y = -\frac{1}{2}x$

C. $y = \frac{1}{2}x$

D. $y = x + 18$

E. $y = 2x$

DO YOUR FIGURING HERE.

$(-12, 6)$ $(0, 0)$

$\frac{0-6}{0-(-12)} = \frac{-6}{12} = -\frac{1}{2}$

34. Which of the following lists the real values of x that make the expression $\frac{x+1}{x^2-2x-3}$ undefined?

F. 0 only

G. 3 only

H. -1, 3 only

J. 1, -3 only

K. 0, 1, -3 only

$x^2 - 2x - 3 = 0$

$x^2 - 2x = 3$

35. How many positive *prime* factors does the number 30 have?

A. 2

B. 3

C. 5

D. 6

E. 7

36. What is the distance, in coordinate units, between the points $(-4,8)$ and $(2,2)$ in the standard (x,y) coordinate plane?

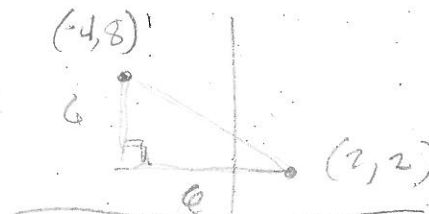
F. $\sqrt{40}$

G. $\sqrt{72}$

H. $\sqrt{104}$

J. $\sqrt{136}$

K. 12



37. The measures of the interior angles of a trapezoid are in the ratio of 3:4:5:6. What is the measure of the largest interior angle?

A. 10°

B. 20°

C. 60°

D. 120°

E. 180°

$3x + 4x + 5x + 6x = 360$

$18x = 360$
 $\frac{18x}{18} = \frac{360}{18}$

38. Which of the following ratios is equivalent to the ratio of 2^2 to 4^4 ?

F. 1 to 2

G. 1 to 4

H. 1 to 8

J. 1 to 16

K. 1 to 32



Use the following information to answer questions 39–41.

DO YOUR FIGURING HERE.

Ms. Martín uses a computer program to produce periodic progress reports for the 10 students in her Advanced Mathematics II class. The report gives a student's scores on tests, quizzes, and homework assignments and the average of all 10 students' scores on each. The progress report shown below was produced near the end of the term for Richard Baldwin.

Student: Baldwin, Richard			
Task	Possible points	Student score	Class average
Homework #1	100	96	86.7
Homework #2	100	85	81.2
Homework #3	100	89	87.4
Quiz #1	100	85	77.9
Quiz #2	100	93	89.1
Quiz #3	100	89	85.1
Quiz #4	100		
Test #1	100	88	81.5
Test #2	100	68	81.0
Pretest	100	69	76.6
Posttest	100		

39. The solution of which of the following inequalities would tell Richard the minimum score (represented by x) he needs to earn on Quiz #4 in order to have a quiz average of at least 90?

A. $\frac{85 + 93 + 89 + x}{4} \geq 90$

B. $\frac{x - (85 + 93 + 89)}{4} \geq 90$

C. $\frac{85 + 93 + 89 + x}{3} \geq 90$

D. $\frac{85 + 93 + 89 + x}{2} \geq 90$

E. $\frac{\frac{85 + 93 + 89}{3} + x}{2} \geq 90$



DO YOUR FIGURING HERE.

40. When Richard took the posttest, his score was 78. The increase from his pretest score to his posttest score was what whole percent of his pretest score?

F. 88%
 G. 87%
 H. 74%
 J. 13%
 K. 12%

41. In Ms. Martín's class, each student who does not turn in a homework assignment receives a score of 0 for the assignment. Of the 10 students in Ms. Martín's Advanced Mathematics II class, what is the maximum possible number who did not turn in Homework #1?

A. 0
 B. 1
 C. 2
 D. 3
 E. Cannot be determined from the given information

42. Which of the following could be the last 2 digits of an integer that is a perfect square?

F. 12
 G. 23
 H. 56
 J. 87
 K. 99

43. Which of the following gives the fractions $-\frac{3}{7}$, $-\frac{2}{3}$, and $-\frac{5}{9}$ in order from least to greatest?

A. $-\frac{5}{9} < -\frac{2}{3} < -\frac{3}{7}$
 B. $-\frac{3}{7} < -\frac{2}{3} < -\frac{5}{9}$
 C. $-\frac{3}{7} < -\frac{5}{9} < -\frac{2}{3}$
 D. $-\frac{2}{3} < -\frac{3}{7} < -\frac{5}{9}$
 E. $-\frac{2}{3} < -\frac{5}{9} < -\frac{3}{7}$

44. What is the midpoint of the line segment with endpoints (3,6) and (10,-24) in the standard (x,y) coordinate plane?

F. $(-\frac{7}{2}, 15)$
 G. $(\frac{7}{2}, -15)$
 H. (5,-12)
 J. $(\frac{13}{2}, -9)$
 K. (13,-18)



45. A circle in the standard (x,y) coordinate plane has center $(3,8)$ and is tangent to the x -axis. The point (x,y) is on the circle if and only if x and y satisfy which of the following equations?

A. $(x - 3)^2 + (y - 8)^2 = 64$
 B. $(x - 3)^2 + (y - 8)^2 = 9$
 C. $(x + 3)^2 + (y + 8)^2 = 64$
 D. $(x + 3)^2 + (y + 8)^2 = 9$
 E. $(x + 3)^2 + (y + 8)^2 = 8$

DO YOUR FIGURING HERE.

46. The circle in the figure below has a radius of 8 centimeters. What is the length, in centimeters, of the arc that is intercepted by a central angle of 60° ?

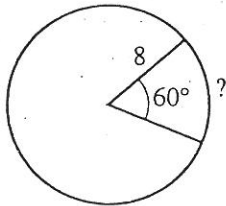
F. $\frac{4\pi}{3}$

G. $\frac{8\pi}{3}$

H. 4π

J. 8π

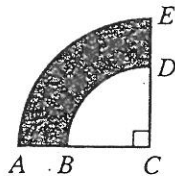
K. 10π



47. If $27 \cdot 3^x = 9^{y-6}$ and $y = 6$, then what is the real number value of x ?

A. -3
 B. -2
 C. -1
 D. 0
 E. 3

48. The diagram below shows a quarter of each of 2 circles both having point C as their center. Point B lies on $\overline{AC}$ and point D lies on $\overline{CE}$. The length of $\overline{BC}$ is 5 inches and the length of $\overline{AB}$ is 3 inches. What is the area, in square inches, of the shaded portion?



F. 4π

G. $\frac{25}{4}\pi$

H. $\frac{39}{4}\pi$

J. $\frac{55}{4}\pi$

K. 16π

www.actexam.net



49. From a blimp 1,130 feet above the ground, the angle of depression to a marker on the ground is 52° , as shown in the figure below. Which of the following expressions gives the distance along the ground, in feet, from the marker to point P directly below the blimp?

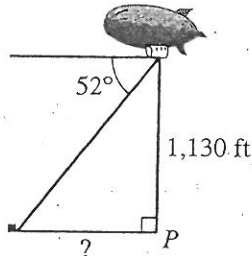
A. $1,130 \tan 38^\circ$

B. $1,130 \sin 38^\circ$

C. $1,130 \cos 38^\circ$

D. $\frac{1,130}{\cos 38^\circ}$

E. $\frac{1,130}{\tan 38^\circ}$



DO YOUR FIGURING HERE.

50. Given a real number n such that $|n| > n$, which of the following *must* be true?

F. $n < 0$

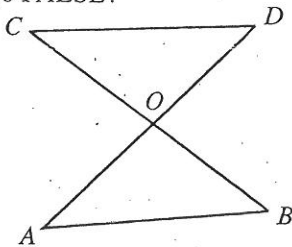
G. $n = 0$

H. $n = -n$

J. $n > |-n|$

K. $n > \sqrt{n}$

51. For all figures in which point O is the intersection of $\overline{AD}$ and $\overline{BC}$, and $m\angle CDO > m\angle BAD$ (the measure of $\angle CDO$ is greater than the measure of $\angle BAD$), such as the figure shown below, which of the following statements is *always* FALSE?



A. $\overline{AB} \parallel \overline{CD}$

B. $\overline{AB} \cong \overline{CD}$

C. $\overline{AD} \perp \overline{BC}$

D. $m\angle BCD > m\angle BAD$

E. $\angle AOB \cong \angle COD$

52. The variables a , b , and c are all integers and $a + b + c = 50$. If $a < 0$ and $0 < b < 30$, then the minimum possible value for c is:

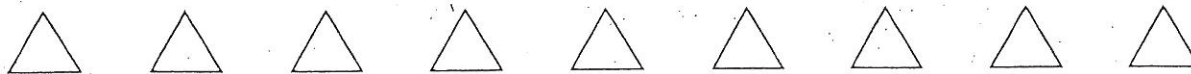
F. 16

G. 17

H. 21

J. 22

K. 25



DO YOUR FIGURING HERE.

53. A polynomial function $P(x)$ has degree n . The graph in the standard (x,y) coordinate plane of $y = P(x)$ contains exactly 3 points on the x -axis. Which of the following could NOT be the value of n ?

A. 6
B. 5
C. 4
D. 3
E. 2

54. The radius of a right circular cone is doubled and its height is tripled to form a new right circular cone. What is the ratio of the volume of the original cone to the volume of the new cone?

(Note: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$ for radius r and height h .)

F. 1:6
G. 1:12
H. 1:18
J. 1:36
K. 1:216

55. Two integer values of x solve the equation $x^2 - x = b$. One of these is positive, and the other is negative. Which of the following could be the value of b ?

A. -20
B. -1
C. 0
D. 9
E. 12

56. Which of the following integers *must* be a factor of the sum of any 3 consecutive positive integers?

F. 2
G. 3
H. 4
J. 5
K. 6

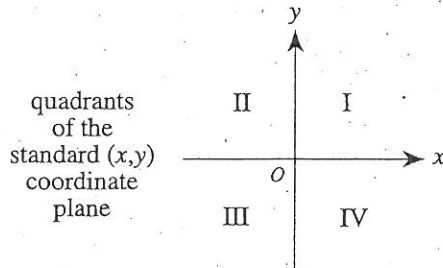
57. A solid block of wood has 6 rectangular faces. The length, width, and height of the block are a , b , and c inches, respectively. Which of the following expressions represents the block's surface area, in square inches?

A. $6abc$
B. $4ab + 2bc$
C. $(a + b + c)^2$
D. $2a + 2b + 2c$
E. $2ab + 2bc$

DO YOUR FIGURING HERE.

58. The quadrants of the standard (x,y) coordinate plane are shown in the figure below. Given that the vertex of $\angle\alpha$ is the origin, the positive x -axis is one side of $\angle\alpha$, its other side lies in Quadrant II, and $\sin \alpha = \frac{\sqrt{3}}{2}$, what is $\cos 2\alpha$?

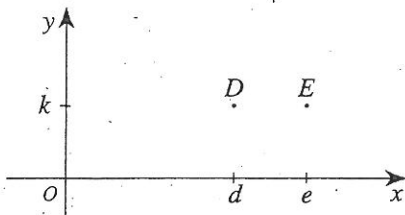
- F. $-\frac{\sqrt{3}}{2}$
- G. $-\frac{1}{2}$
- H. $\frac{1}{4}$
- J. $\frac{1}{2}$
- K. $\frac{\sqrt{3}}{2}$



59. For a specific nonzero value of b , the graph of $y = \sin bx$ in the standard (x,y) coordinate plane intersects the x -axis at k values of x in the interval $0 \leq x < 2\pi$. At how many values of x in this interval does the graph of $y = 2 \sin bx$ intersect the x -axis?

- A. k
- B. $k - 2$
- C. $k + 2$
- D. $\frac{k}{2}$
- E. $2k$

60. In the standard (x,y) coordinate plane below, the coordinates of D are (d,k) , and the coordinates of E are (e,k) . When D is rotated 180° about E , the result is F . In terms of d , e , and k , what are the coordinates of F ?



- F. $(e + d, 2k)$
- G. $(e - d, 0)$
- H. $(e - d, k)$
- J. $(2e - d, k)$
- K. $(2d - e, k)$

END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are four passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

PROSE FICTION: This passage is adapted from the novel *Agassiz* by Sandra Birdsell (©1991 by Sandra Birdsell).

My grandmother Omah supplies flowers for weddings and funerals. In winter, the flowers come from the greenhouse she keeps warm with a woodstove as long as she can; and then the potted begonias and asters
5 are moved to the house and line the shelves in front of the large triple-pane window. She has no telephone. I am the messenger. I bring requests to her, riding my bicycle along the dirt road to her cottage.

A dozen or two glads please, the note says. The
10 bride has chosen coral for the color of her wedding and Omah adds a few white ones because she says that white is important at a wedding. She does not charge for this service. It is unthinkable to her to ask for money to do this thing which she loves.

She has studied carefully the long rows of blossoms to find perfect ones with just the correct number of buds near the top, and laid them gently on newspaper. She straightens and absently brushes perspiration from her brow. She frowns at the plum tree in the
20 corner of the garden where the flies hover in the heat waves.

"In Russia," she says as she once more bends to her task, "we made jam. Wild plum jam to put into fruit pockets and platz." Her hands, brown and earth-stained,
25 feel for the proper place to cut into the last gladiolus stalk. "What now is a plum tree but a blessing to the red ants and flies only?" While she steps into the house, I search the ground beneath the tree to try to find out what offends her so greatly. I can see red ants crawling
30 over sticky, pink pulp, studying the dynamics of moving one rotting plum. I recall her words as I pedal back towards the town with the glads in my wire basket.

Laurence is waiting for me at the river.

35 "Drat, wild plums, that's just what I need," Laurence says and begins pacing up and down the river bank.

"What do you need plums for?"

"Can't you see," he says. "Wild plums are perfect
40 to make wine." I wonder at the tone of his voice when it is just the two of us fishing. He has told me two bobbers today instead of one and the depth of the stick must be screwed down into the muck just so. Only he can do it. And I never question as I would want to
45 because I am grateful for the world he has opened up to me.

I have loved you since grade three, my eyes keep telling him. I defy my father and play with you anyway. It is of no consequence to Laurence that daily our
50 friendship drives wedges into my life.

"Come with me." I give him my hand.

Omah bends over in the garden picking tomatoes, her instep swollen, mottled blue with broken vessels, her beautiful white hair hidden beneath a kerchief.

55 Laurence stands away from us, his arms folded across his chest as though he were bracing himself against extreme cold.

"His mother could use the plums," I tell Omah. Her eyes brighten and she half-runs to her house. When
60 she returns with pails for picking, Laurence's arms hang down by his sides.

"You tell your Mama," she says to Laurence, "that it takes one cup of sugar to one cup of juice for the jelly."

65 I beg Laurence silently to be polite in front of her. Her freckled hands pluck fruit joyfully.

"In the old country, we didn't waste fruit. Not like here where people let it fall to the ground and then go to the store and buy what they could have made for
70 themselves."

Laurence has sniffed out my uneasiness. "My mother makes good crabapple jelly."

She studies him with renewed interest. When we each have a pail full of the fruit, she tops it with several
75 of the largest unblemished tomatoes I have ever seen.

We leave her standing at the edge of the road waving. I am so proud that I want to tell Laurence

about the apple that is named for her. She had experimented with crabapple trees for years and in recognition of her work, the experimental farm has given a new apple tree her name.

When we are well down the road we stop to rest. I sit and chew the tender end of a foxtail. Laurence chooses the largest of the tomatoes carefully, and then, his arm a wide arc, smashes it against a telephone pole.

I watch red juice dripping against the splintered gray wood. The sun is dying. It paints the water tower shades of gold. The crickets in the ditch speak to me of Omah's greenhouse where they hide behind earthenware pots.

What does Laurence know of hauling pails of water from the river, bending and trailing moisture, row upon row? What does he know of coaxing seedlings to grow?

I turn from him and walk with my face reflecting the fired sky and my dust-coated bare feet raising puffs of anger in the fine warm silt.

"Hey, where are you going?" Laurence calls to my retreating back. "What did I do?"

- Which of the following statements accurately summarizes the central emotional shift that takes place in this passage?
 - A young narrator comes to regret her role as the messenger for her grandmother's business because it leads to personal disappointment.
 - A grown woman feels her enthusiasm for gardening decline in the face of the disrespect she is shown by a young man.
 - A young narrator becomes more attached to her grandmother and less attached to the boy she loves as a result of trying to please both of them.
 - A young boy who is trying to please the narrator is overwhelmed with resentment when he realizes his efforts are misunderstood.
- Laurence's approach to fishing and Omah's approach to gardening reveal that each has:
 - strong opinions about preferred techniques in their areas of expertise.
 - an indifferent attitude toward carrying out familiar tasks.
 - a sense of responsibility to uphold traditions established by others.
 - little tolerance for the demands made upon them by the larger community.
- The passage states that the narrator's feelings for Laurence include all of the following EXCEPT:
 - gratitude.
 - love.
 - envy.
 - anger.

- It can reasonably be inferred from the passage that bringing Laurence and Omah together takes an emotional toll on the narrator because:
 - Laurence damages the plums that Omah has carefully set aside for making jelly.
 - Laurence asks questions that require Omah to defend her values.
 - Omah insults Laurence by her reluctance to offer his mother plums to make jelly.
 - she has strong attachments to both and fears that one might hurt or offend the other.
- The author most likely provides images contrasting Omah and Laurence in the twelfth and thirteenth paragraphs (lines 55–61) to help the reader:
 - visualize physical details that reveal Omah and Laurence's opposing attitudes.
 - imagine vividly the characteristics of Omah's house and land in contrast to Laurence's home.
 - sympathize with the narrator, who is glad that Laurence's mother can use the plums.
 - understand the disagreement between Omah and Laurence as they pick plums together.
- Laurence's behavior toward Omah during his visit to the garden can best be characterized as:
 - initially aloof then tentatively involved.
 - clumsy but well-meaning in ways that amuse Omah.
 - initially outgoing then increasingly subdued.
 - jumpy until Omah's instructions calm him down.
- The questions posed by the narrator in the twenty-second paragraph (lines 91–93) most strongly indicate that she is:
 - curious about how much Laurence has learned about tending a garden like Omah's.
 - mad at Laurence for demonstrating that he has no respect for Omah's hard work in the garden.
 - disappointed in herself for not explaining to Laurence the significance of Omah's labors.
 - determined to bridge the gap that has opened up unexpectedly between Laurence and Omah.
- It can reasonably be inferred from the first two paragraphs (lines 1–14) that in providing her services to the community, Omah considers it her duty to:
 - fill a customer's order as directed, adhering strictly to the customer's request.
 - adjust an order to conform with her sense of what an occasion calls for.
 - pick flowers from the tops of the plants where the colors of the blooms are the most vivid.
 - ask the narrator detailed questions about the occasion and the customer's preferences.
- As it is used in line 50, the word *wedges* most nearly means:
 - supports.
 - braces.
 - interventions.
 - divisions.

10. It can reasonably be inferred that the narrator's remark "His mother could use the plums" (line 58) is the narrator's way of influencing the attitude of:

F. Laurence's mother toward Omah.
 G. Laurence toward the plums.
 H. Omah toward Laurence.
 J. the narrator's mother toward Laurence's mother.

Passage II

SOCIAL SCIENCE: This passage is adapted from the book *Thurgood Marshall: American Revolutionary* by Juan Williams (©1998 by Juan Williams).

Thurgood Marshall's lifework made him one of America's leading radicals. As a suit-and-tie lawyer, however, he was the unlikely leading actor in creating social change in the United States in the twentieth century. His great achievement was to expand rights for individual Americans. But he especially succeeded in creating new protections under law for America's women, children, prisoners, homeless, minorities, and immigrants. Their greater claim to full citizenship in the Republic over the last century can be directly traced to Marshall. Even the American press has Marshall to thank for an expansion of its liberties during the century.

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But for black Americans especially, Marshall stood as a colossus. He guided a formerly enslaved people along the road to equal rights. Oddly, of the three leading black liberators of twentieth-century America—Thurgood Marshall, Martin Luther King, Jr., and Malcolm X—Marshall was the least well known. Dr. King gained fame as the inspiring advocate of non-violence and mass protests. Malcolm X was the defiant black nationalist whose preachings about separatism and armed revolution were the other side of King's appeals for racial peace. But the third man in this black triumvirate stood as the one with the biggest impact on American race relations.

It was Marshall who ended legal segregation in the United States. He won Supreme Court victories breaking the color line in housing, transportation, and voting, all of which overturned the "separate-but-equal" apartheid of American life in the first half of the century.

It was Marshall who won the most important legal case of the century, *Brown v. Board of Education*, ending the legal separation of black and white children in public schools. The success of the *Brown* case sparked the 1960s civil rights movement, led to the increased number of black high school and college graduates and the incredible rise of the black middle class in both numbers and political power in the second half of the century.

And it was Marshall, as the nation's first African-American Supreme Court justice, who promoted affir-

mative action—preferences, set-asides, and other race-conscious policies—as the remedy for the damage remaining from the nation's history of slavery and racial bias. Justice Marshall gave a clear signal that while legal discrimination had ended, there was more to be done to advance educational opportunity for blacks and to bridge the wide canyon of economic inequity between blacks and whites.

Marshall was busy in the nation's courtrooms, winning permanent changes in the rock-hard laws of segregation. He created a new legal landscape, where racial equality was an accepted principle. He worked in behalf of black Americans but built a structure of individual rights that became the cornerstone of protections for all Americans. Marshall's triumphs led black people to speak of him in biblical terms of salvation: "He brought us the Constitution as a document like Moses brought his people the Ten Commandments," the NAACP board member Juanita Jackson Mitchell once said.

The key to Marshall's work was his conviction that integration—and only integration—would allow equal rights under the law to take hold. Once individual rights were accepted, in Marshall's mind, blacks and whites could rise or fall based on their own ability.

Marshall's deep faith in the power of racial integration came out of a middle-class black perspective in turn-of-the-century Baltimore. He was the child of an activist black community that had established its own schools and fought for equal rights from the time of the Civil War. His own family, of an interracial background, had been at the forefront of demands by Baltimore blacks for equal treatment. Out of that unique family and city was born Thurgood Marshall, the architect of American race relations in the twentieth century.

Thurgood Marshall lived to be eighty-four. He held high public office, but for those last thirty years of his life Marshall was reclusive, making few public appearances and rarely talking with reporters. The public knew him primarily as a distant figure whose voice was heard only in the legislative language of Supreme Court dissents. The combination of his reclusiveness and his standing in popular culture as an elderly, establishment figure blinded much of the nation to the importance of his legacy. However, given that Marshall laid the foundation for today's racial landscape, his grand design of how race relations best work makes his life story essential for anyone delving into the subject.

11. According to the passage, in his work to improve race relations, Marshall supported all of the following EXCEPT:

A. affirmative action policies.
 B. the legal process.
 C. community activism.
 D. creating new Supreme Court powers.

12. Based on the first paragraph, which of the following statements would the author make about the focus of Marshall's legal efforts?
- F. As a suit-and-tie lawyer, Marshall focused on those who had the greatest claim to citizenship.
 - G. Marshall was a radical who focused on the freedoms of the press in order to publicize his civil rights work.
 - H. Marshall worked to protect the individual rights of all Americans, focusing on those who had been denied full protection of the law.
 - J. Marshall's focus on social change created new rights for prisoners, children, and the homeless, who until then were not citizens of the Republic.
13. It can reasonably be inferred that the author refers to Marshall as a "colossus" (line 15) in order to illustrate that:
- A. within the African American community, Marshall's guidance was similar to that of Malcolm X and King.
 - B. by virtue of his work, Marshall was a monumental figure, especially to those in the African American community.
 - C. for Marshall to have won so many important legal cases, his legal training must have been colossal.
 - D. though Marshall's work has endured through time, it is little known to the general public.
14. The passage states that Marshall's legal victory in the landmark case *Brown v. Board of Education* paved the way for:
- F. the abrupt end of the civil rights movement.
 - G. the significant rise of the African American middle class.
 - H. increased suits against other boards of education.
 - J. Marshall's triumphant retirement from the Supreme Court.
15. Which of the following most accurately states the main idea of the fifth paragraph (lines 42–51)?
- A. Marshall questioned the effectiveness of laws created to remedy damage done by the nation's history of racial inequity.
 - B. As the nation's first African American Supreme Court justice, Marshall signaled that legal discrimination had ended.
 - C. Marshall believed that there must be an ongoing commitment to providing increased educational and economic opportunity as well as legal equality for African Americans.
 - D. Marshall believed affirmative action and other race-conscious policies would not remedy the wide canyon of inequity found in the United States.
16. The passage indicates that all of the following elements contributed to Marshall's belief in racial integration EXCEPT:
- F. his family's middle-class status.
 - G. the interracial background of his family.
 - H. his exposure to an activist African American community.
 - J. the long time he was in public office.
17. It can reasonably be inferred from the last paragraph that Marshall's reclusive nature in his later years had which of the following effects?
- A. It undermined the validity of previous legal arguments he made before the Supreme Court.
 - B. It increased the nation's curiosity and his respected standing as a part of popular culture.
 - C. It heightened interest in his radical approach to race relations in the African American community.
 - D. It dimmed public understanding of the magnitude of his lifelong accomplishments.
18. According to the author, as the third man in the "black triumvirate," Marshall was:
- F. a supporter of both Malcolm X's separatist views and King's appeals for racial peace.
 - G. closer to the radical beliefs of Malcolm X than to King's.
 - H. the most famous of the twentieth-century "black liberators."
 - J. ultimately responsible for having the greatest impact on race relations.
19. According to the passage, Marshall believed that equal rights for all citizens could not be fully achieved without:
- A. a new legislative language.
 - B. nonviolent mass protests.
 - C. full racial integration.
 - D. race-neutral policies.
20. In the last paragraph, the author indicates that studying Marshall's lifework is essential to understanding which of the following?
- F. The development of race relations in the United States in the twentieth century
 - G. The roles of Malcolm X and King as African American liberators
 - H. The sophisticated legal definition of integration
 - J. The legislative language of the Supreme Court

Passage III

HUMANITIES: This passage is adapted from "Borderline" by Kathleen Norris (©1995 by Kathleen Norris).

Since my book *Dakota* became a surprise best-seller, I've received thousands of letters responding to it. Along with the reviews, these responses have been, for the most part, positive, and often so enthusiastic that I wonder if the person has read the same book that I wrote. I was enormously relieved, then, to receive the following from my sister Becky: "How are you? Everyone here talks about you and your book. I feel left out. I tried to read it, but it was boring."

Not for the first time in our forty-two-year relationship, my sister Becky had acted as a kind of *amma* for me, a desert mother who challenges my complacency, and allows me to see the world (and myself) in a new light. I was reminded of a saying by Amma Synclectica, a desert monastic of fourth-century Egypt: "Just as it is impossible to be at the same moment a plant and a seed, so it is impossible for us to be surrounded by worldly honor and at the same time to bear heavenly fruit."

Synclectica sums up, I believe, the difficulty writers have in America in surviving success: to keep bearing fruit, one must keep returning, humbly, to the blank page and not pay much heed to the "noted author" the world wants you to be. Becky's letter was a godsend. Reading it over, I laughed myself silly, and found myself released from much of the tension induced by sudden notoriety and too much literary hoo-ha.

Becky's life has been a kind of desert. When she was born, the doctors gave my mother too strong a dose of sedative. Having already given birth to two children, she knew something was wrong. Precious oxygen was lost. My sister's brain was irreversibly damaged.

Becky is diagnosed as "borderline." She is intelligent enough to comprehend what happened to her when she was born. She is not intelligent enough to do mathematical computation. A tutor my parents hired when Becky was about ten years old told us that Becky could grasp a concept long enough to work out several problems in the course of an hour-long session, but that by the next week she'd have forgotten what she'd learned and have to start from scratch. Her elementary schools were just passing her along; there were no "special ed" programs then, and no one knew what to do with her, where she belonged. Becky's life has been lonely in ways that most of us could not comprehend.

Our parents decided when Becky was very young that she didn't belong in an institution, but with us. I believe that being raised with myself and a brother, both older, and one younger sister was good for Becky; I know it was good for me. Very early on, I had to learn to respect Becky's intelligence, although it was very different from mine. I also came to respect her tenacity.

When I was in high school I began to discover how much my sister and I had in common. We were both in difficult situations—I was a shy, ungainly newcomer at a school where many of the students had been together since kindergarten, and Becky had a particularly unsympathetic teacher. On coming home from school, she'd immediately go to her room and play rock music while she danced around her room (and sometimes on her bed). She talked to herself, incessantly and loudly. The family accepted all this as something Becky needed to do.

One day, as Becky carried on her usual "conversations," with her teacher, with other girls in her class, with a boy who'd made fun of her, I was doing homework in the room next door and realized that I, too, needed release from daily tensions, a way to daydream through the failed encounters and make them come out right. Usually, I lost myself in reading or practicing the flute, but sometimes I listened to music and imagined great careers for myself, great travels, great loves. I didn't have the nerve to stomp around my room and yell, as my sister was doing, but our needs were the same.

We were both struggling with our otherness, although I suspect I did not know this then. Now that I'm a writer, it's clear to me. Rejection comes to everyone, of course, but for those who are markedly different from their peers, it is a daily reminder of that difference. To most people, my sister and I didn't seem to have much in common; but I knew from that day on that we were remarkably alike.

If nothing else, this insight helped me to survive my new school. I knew that getting a C was not the worst thing in the world. And when I got an A, I could put it in perspective. I knew there were other kinds of intelligence that were just as valuable; needs that could not be satisfied in school.

21. One of the main points the narrator makes in this passage is that:

- A. rejection is a normal part of high school life.
- B. there should be "special education" programs in every school.
- C. she and her sister had a similar need for acceptance.
- D. nearly everyone has an *amma* to give him or her advice.

22. As depicted in the passage, the narrator's attitude toward literary success is best described by which of the following statements?
- F. Receiving praise for literary success indicates creativity and is a worthy goal to pursue.
 - G. The opinions of family and friends are helpful to authors but not a true measure of literary success.
 - H. Receiving praise for literary success can have a negative effect on one's writing.
 - J. Noted authors deserve the public's respect because they endure considerable hardship in order to be successful.
23. The narrator's reaction to Becky's assessment of the book *Dakota* reveals that the narrator:
- A. wishes she had Becky's love.
 - B. did not understand Becky's critique.
 - C. has a healthy sense of humor.
 - D. does not agree with Becky.
24. As it is used in line 77, the word *otherness* refers to the narrator's and Becky's:
- F. fondness for each other.
 - G. foreign origin.
 - H. need to be alone.
 - J. feelings of isolation.
25. The narrator claims that Becky "acted as a kind of *amma*" (line 11) because:
- A. Becky's criticism allowed the narrator to reassess herself at a time when that was what she needed.
 - B. the role Becky most often assumed in the family was as a spiritual guide during times of crisis.
 - C. after reading Becky's assessment of her book, the narrator was determined to write books Becky would like.
 - D. unlike most readers, both Becky and Amma Synclectica found the book *Dakota* boring.
26. In the saying quoted in lines 16–19, Amma Synclectica advises people to be:
- F. humble.
 - G. tolerant.
 - H. fair.
 - J. intelligent.
27. It can most reasonably be inferred that the narrator's own version of bearing what Amma Synclectica called "heavenly fruit" (line 19) would be to:
- A. follow her instincts above all.
 - B. write high-quality literature.
 - C. be accepted and honored by her peers.
 - D. continue to be praised by literary reviewers.
28. It can most reasonably be inferred that the narrator compares Becky's life to a desert in line 29 as a way of describing Becky's:
- F. need for honesty.
 - G. dry sense of humor.
 - H. consistent hopefulness.
 - J. ongoing loneliness.
29. The sixth paragraph (lines 47–53) reveals that the narrator is:
- A. thoughtful, naive, and nostalgic.
 - B. secure, affectionate, and ironic.
 - C. open-minded, accepting, and grateful.
 - D. aware, respectful, and tenacious.
30. Becky's method of coping with difficulty was similar to the narrator's method in that both girls:
- F. carried on pretend conversations with each other.
 - G. acted unconcerned about what others thought of them.
 - H. relied on their parents and siblings for advice.
 - J. used their imagination to make themselves feel better.

Passage IV

NATURAL SCIENCE: This passage is adapted from "Were Dinosaurs Cold Blooded?" by Brendan J. Koerner (©1997 by U.S. News and World Report Inc.). Warm-blooded creatures maintain a nearly constant body temperature under most conditions.

The claim that dinosaurs were warm blooded has divided the paleontological world into rival camps. For some, endothermy, the scientific name for warm bloodedness, is the only way to explain the dinosaurs' evolutionary success. Without the ability to keep their bodies at optimum temperatures regardless of their surroundings, they argue, dinosaurs could never have dominated the globe for 160 million years. Skeptics counter that ectothermy, the proper label for cold bloodedness, was the logical strategy for dinosaurs living in the Mesozoic Era's generally sweltering heat—and, this group claims, the only option that is supported by physiological, rather than circumstantial, evidence.

The revisionist view that has so captured the public imagination has long been led by Robert Bakker, a crusader against slow-moving, dimwitted, crocodilian dinosaurs. He painted a picture of dinosaurs that were every bit as endothermic as humans. Instead of spending their days lazily basking in the sun and occasionally trudging along at a torpid pace, Bakker's dinosaurs moved at constant speeds, their postures fully erect in the manner of birds and mammals. "Meat-eating dinosaurs related to *Tyrannosaurus rex* cruised at 3 to 4 miles an hour," claims Bakker, who bases his conclusion on fossilized footprints. "No turtle anywhere cruises at 3 to 4 miles an hour."

Bakker and his acolytes also point to dinosaurs' relatively fast growth as evidence of endothermy. Mammals and birds, which develop quickly compared with ectothermic reptiles, have bones characterized by microscopic channels that appear complex and crystal-like under the microscope. These elegant patterns form when growing bone meets and meshes with connective tissue, capturing blood vessels in dense, woven structures called Haversian canals. Armand de Ricqlès, a University of Paris anatomist, found that dinosaur bones exhibited those same intricate channels rather than the simpler, less dense structures common to reptiles. "We see the same well-vascularized bone in mammals but not in turtles and crocodiles," says Kevin Padian, a paleontologist at the University of California-Berkeley. "The way the bones grew, dinosaurs seem to have been active all the time." That pace of activity, argue Bakker and his cohorts, is the telltale sign of warm bloodedness.

Many within the scientific community remain wary of Bakker's claims. Since measurements show that endotherms require up to 20 times more food than ectotherms, some question how the gigantic dinosaurs could possibly have eaten enough if they were warm blooded.

The evidence based on bone structures has come under fire, too. Tomasz Owerkowicz, a Harvard Uni-

versity researcher, has asserted that the dense canals that de Ricqlès detected could have resulted from physical exertion rather than endothermy. In an ingenious experiment, Owerkowicz gave cold-blooded monitor lizards regular treadmill workouts and then compared their bones with those of nonaerobically conditioned contemporaries. The well-exercised group showed the same kind of complex channels characteristic of mammals, birds, and de Ricqlès's dinosaurs, suggesting that Haversian canals are causally linked to an active lifestyle rather than warm bloodedness. South African histologist Anusuya Chinsamy has also countered some of the bone structure argument, contending that dinosaur bones exhibit bands called lines of arrested growth. These are characteristic of modern-day ectotherms, whose growth rate speeds up and slows down according to seasonal temperature fluctuations. Chinsamy concluded that dinosaurs grew at a more reptilian pace than envisioned by the Bakkerites.

The ectothermic side has sought to boost its case with hard physiological evidence. Turbinates, tiny whisks of bone or cartilage deep inside the nasal cavities of mammals and birds, make warm bloodedness possible by limiting water loss. Turbinates have never been found in living ectotherms—nor in dinosaurs.

Many on the cold-blooded side now use the term "gigantothermy" to describe the unique energetics of large dinosaurs. Being huge is one way to maintain a relatively constant body temperature despite cold bloodedness: Large things—which have a lot of bulk in relation to their skin area—lose heat to the outside world much more slowly than do small things. If dinosaurs were indeed cold blooded, the slow heat loss associated with gigantothermy would allow them to stay relatively warm—and avoid a reptilian torpor—when confronted by the night or an overcast day.

31. The function of the information provided in the passage's last four paragraphs (lines 46–89) is primarily to:
 - A. provide a counterpoint to the theories advanced by Bakker and his revisionist-minded colleagues.
 - B. disclose information about turbinates that undermines the theories put forward by Padian and Bakker.
 - C. show that Haversian canals have a wholly different purpose and origin than de Ricqlès thought.
 - D. make clear the importance of the research done by Chinsamy and Owerkowicz.
32. Which of the following questions best addresses the main point of the first paragraph?
 - F. Why were dinosaurs able to dominate the globe for 160 million years?
 - G. When did the paleontological world divide into two rival groups?
 - H. Why are some paleontologists skeptical about ectothermy?
 - J. Were dinosaurs endothermic or ectothermic?

33. It can reasonably be inferred that the word *acolytes*, as it is used in line 27, refers to paleontologists who:
- A. disagree with Bakker's theories.
 - B. share Bakker's revisionist views.
 - C. have mixed feelings about Bakker's work.
 - D. think dinosaurs were ectothermic.
34. It is clear from the third paragraph (lines 27–45) that Haversian canals can only be viewed:
- F. by scientists.
 - G. in dinosaurs.
 - H. by Bakkerites.
 - J. through a microscope.
35. It can reasonably be inferred from the fifth paragraph (lines 52–72) that the description of Owerkowicz's work functions in the passage as a whole to:
- A. demonstrate that de Ricqlès did not find Haversian canals in the dinosaurs he studied.
 - B. present an alternative explanation of how the dinosaurs' Haversian canals might have developed.
 - C. explain what caused ectothermic dinosaurs to evolve differently from endothermic dinosaurs.
 - D. illustrate the way in which reptiles could be made to perform in ingenious experiments.
36. The research by Chinsamy cited in lines 64–72 serves which of the following purposes?
- F. It enhances the findings of de Ricqlès.
 - G. It contradicts the theories of Owerkowicz.
 - H. It lends support to the theories of Owerkowicz.
 - J. It augments the theories of Bakker and his associates.
37. The fact cited in lines 77–78 is most consistent with the ideas about dinosaurs attributed to which of the following scientists?
- A. Chinsamy
 - B. Padian
 - C. Bakker
 - D. de Ricqlès
38. The passage states that the growth rate of modern ectotherms is directly related to which of the following?
- F. Their food
 - G. Habitat plants
 - H. Seasonal temperature
 - J. Their size
39. The sixth paragraph (lines 73–78) indicates that turbinates are found in all of the following EXCEPT:
- A. birds.
 - B. mammals.
 - C. endotherms.
 - D. ectotherms.
40. As it is used in the last paragraph, the term *gigantothermy* most nearly means that the:
- F. larger the dinosaur, the easier it is for the dinosaur to be endothermic.
 - G. larger the dinosaur, the easier it is for the dinosaur to retain body heat.
 - H. smaller the dinosaur, the more likely that that dinosaur will be ectothermic.
 - J. smaller the dinosaur, the more likely it will exist in a perpetual state of reptilian torpor.

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.
DO NOT RETURN TO A PREVIOUS TEST.



SCIENCE TEST

35 Minutes—40 Questions

DIRECTIONS: There are seven passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage 1

Since 1880, Earth's average surface temperature has increased by 0.5°C . This *global warming* has caused ocean surface temperature increases and the melting of parts of the polar ice caps. Two scientists discuss the cause of these events.

Scientist 1

Global warming is caused by human activities that have significantly increased the atmospheric concentration of CO_2 above the concentration prior to 1880. CO_2 is one of several *greenhouse gases* (gases that trap heat emitted from Earth's surface). Human activities, primarily the burning of fossil fuels, have added significant additional CO_2 to the air (6 billion tons per year at present). This has increased the atmospheric CO_2 concentration by 30% since 1880. The higher the atmospheric CO_2 concentration, the warmer Earth's average surface temperature. While atmospheric CO_2 concentrations have been steadily increasing over the past 18,000 years, the annual rate of increase over the past 100 years has been 10 to 100 times faster than the rate of increase prior to 1880. Eight of the warmest years on record have occurred since 1979.

Scientist 2

Changes in atmospheric CO_2 concentration have little or no effect on average surface temperature. Global warming is caused by an increase in the energy output of the Sun and by regular changes in the shape of Earth's orbit. A less elliptical orbit corresponds to a warmer climate. These 2 factors explain at least 80% of the observed surface temperature increase. The rest can be explained by decreases in the amount of atmospheric sulfate aerosols and increases in the stratospheric ozone content. The 6 billion tons/yr of CO_2 from human activities is a very small proportion of the total 186 billion tons/yr of CO_2 that enter the atmosphere. Studies of climate records over hundreds of thousands of years show that average surface temperature has increased and decreased in regular cycles that coincide with increases and decreases in solar energy output. This parallel relationship is particularly evident in the steady warming of average surface temperature since the end of the last ice age 18,000 years ago.

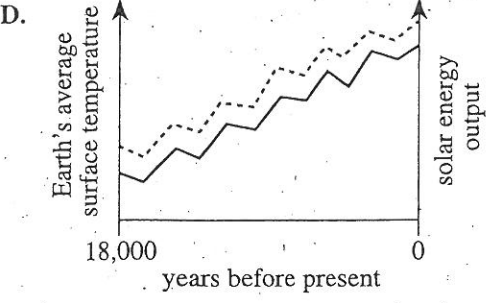
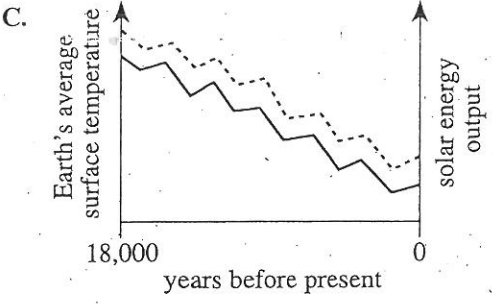
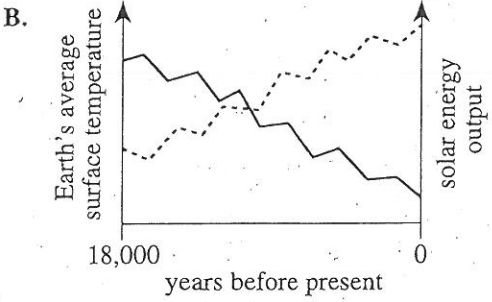
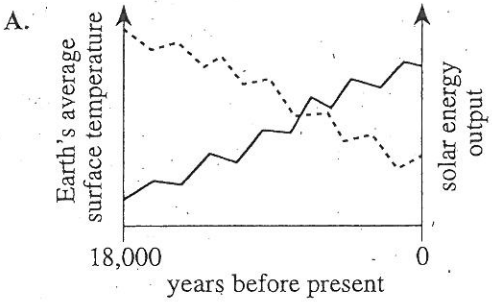
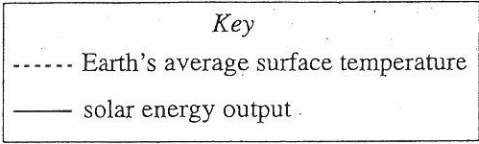
1. If Earth's average surface temperature continues to increase, both Scientist 1 and Scientist 2 would most likely agree that which of the following environmental changes would occur?

A. Earth's polar ice caps would continue to increase in volume.
B. Earth's polar ice caps would continue to decrease in volume.
C. Average ocean temperatures would continue to decrease.
D. Average ocean temperatures would remain constant.

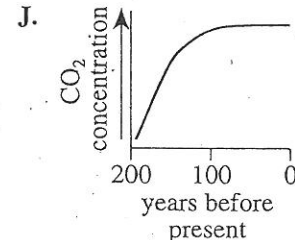
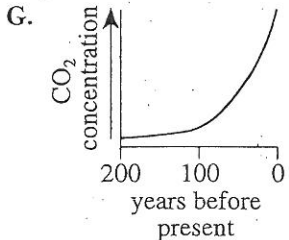
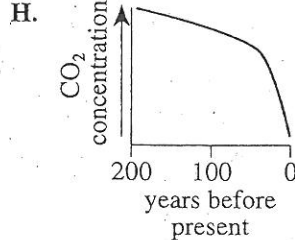
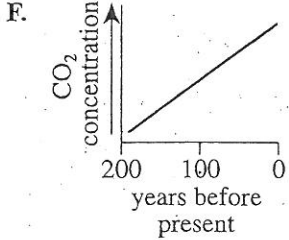
2. If the CO_2 input from human activities could be reduced to zero, Scientist 2 would most likely state that this change would have which of the following resulting effects, if any, on global climate? This change would:

F. cause an immediate increase in global temperature.
G. cause an immediate decrease in global temperature.
H. have no effect on the global climate.
J. accelerate the breakdown of the ozone layer.

3. If Scientist 2 is correct, which of the following graphs would best represent the relationship between solar energy output and Earth's average surface temperature trends over the past 18,000 years?



4. According to the viewpoint of Scientist 1, which of the following graphs best represents the increase in atmospheric CO₂ concentration over the past 200 years?



5. Scientist 2 would most likely agree with which of the following statements about the solar energy output 18,000 years ago, at the end of the last ice age? Eighteen thousand years ago, solar energy output was:

- A. nearly zero.
- B. higher than it is today.
- C. the same as it is today.
- D. lower than it is today.

6. Suppose ozone is a greenhouse gas produced by human activities. Scientist 1 would most likely agree that if increases in stratospheric ozone content contribute to the current global warming, these increases do so because ozone:

- F. can be changed to oxygen through chemical reactions in the atmosphere.
- G. has the ability to trap heat that would otherwise be lost to space.
- H. is the major gaseous component of Earth's atmosphere.
- J. has a molecular mass much lower than the molecular mass of CO₂.

7. Scientist 2 would most likely agree that, compared to the amount of CO₂ that human activities put into the atmosphere each year, the amount put into the atmosphere each year from sources other than human activities is how many times greater?

- A. 5 times greater
- B. 10 times greater
- C. 20 times greater
- D. 30 times greater

Passage II

Crude oil contains thousands of chemical compounds. To obtain commercially useful compounds, crude oil is distilled in a *fractionating tower* (see Figure 1). The crude oil is first heated to 400°C in a furnace, then pumped into the fractionating tower. As vapors from the oil rise, they cool and condense. *Fractions* are collected at different heights in the tower. Each fraction contains a mixture of hydrocarbon molecules with similar boiling points. Table 1 shows information about some of the fractions that can be collected. Figure 2 shows the composition of crude oils obtained in 6 different geographic locations.

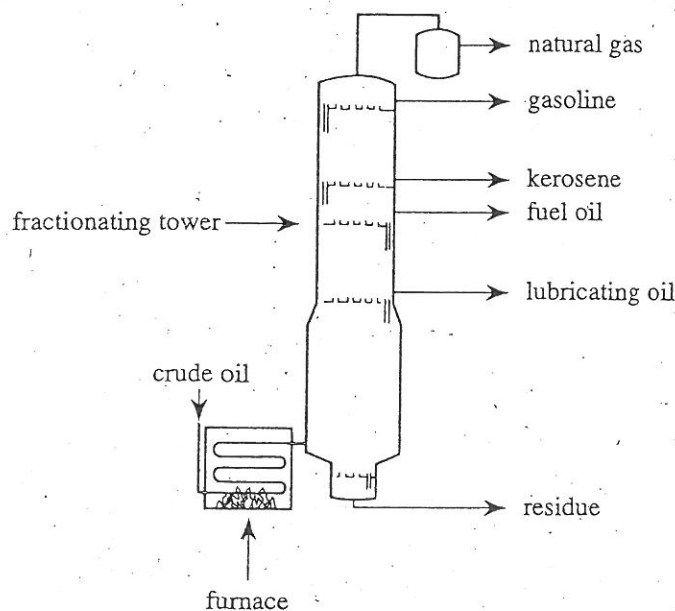


Figure 1

Figure 1 adapted from American Chemical Society, *ChemCom: Chemistry in the Community*, 2nd ed. ©1993 by American Chemical Society.

Table 1

Fraction	Typical number of carbon atoms per molecule	Approximate boiling range (°C)
Natural gas	1-4*	< 20
Gasoline	5-12	40-200
Kerosene	12-16	200-300
Fuel oil	15-18	250-350
Lubricating oil	16-24	300-400

*This range indicates that natural gas typically contains a mixture of different hydrocarbon molecules ranging in size from 1 to 4 carbon atoms.

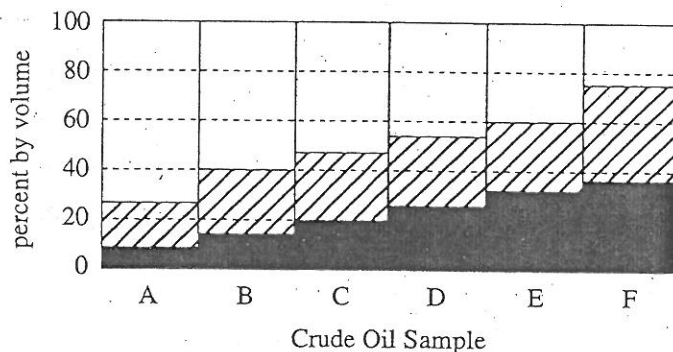
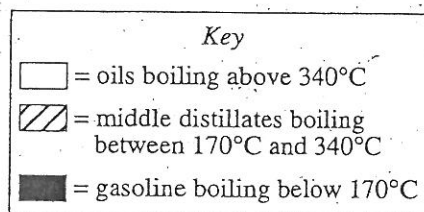


Figure 2

Figure 2 adapted from *The New Encyclopaedia Britannica*. ©1986 by Encyclopaedia Britannica, Inc.

8. According to the information in Table 1, hydrocarbon molecules each having 14 carbon atoms would most likely be found in which of the following fractions?

F. Gasoline
G. Kerosene
H. Fuel oil
J. Lubricating oil

9. Based on the data in Table 1, a fraction containing a mixture of hydrocarbons ranging in size from 8 to 14 carbon atoms per molecule would most likely have a boiling range:

A. below 20°C.
B. between 20°C and 100°C.
C. between 100°C and 250°C.
D. above 250°C.



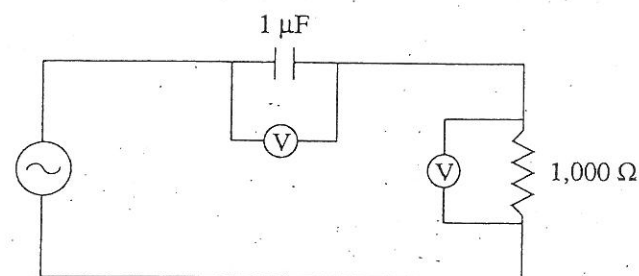
10. Based on the information in Figure 1 and Table 1, a fraction containing a hydrocarbon that has 7 carbon atoms per molecule would most likely be collected:
- F. near the top of the fractionating tower.
 - G. in the middle of the fractionating tower.
 - H. near the bottom of the fractionating tower.
 - J. in the furnace.
11. Based on the information in Figure 1 and Table 1, hydrocarbons found in the residue most likely contain:
- A. 1 to 6 carbons per molecule.
 - B. 7 to 12 carbons per molecule.
 - C. 13 to 18 carbons per molecule.
 - D. more than 18 carbons per molecule.
12. Is the statement "Oils boiling at temperatures above 340°C make up the largest proportion of each of the 6 crude oils shown in Figure 2" supported by Figure 2 ?
- F. Yes; oils boiling above 340°C make up more than 20% of the 6 crude oils.
 - G. Yes; gasoline makes up the smallest proportion of the 6 crude oils.
 - H. No; oils boiling above 340°C make up the smallest proportion of Crude Oil Sample F.
 - J. No; gasoline and middle distillates make up more than 70% of Crude Oil Sample A.

Passage III

An *RC circuit* contains a resistor, a capacitor, and an input voltage source. Some students studied RC circuits containing a source of AC (alternating current) voltage. *Frequency*, in hertz (Hz), is related to the number of times/sec the highest voltage is attained. The measured input voltage from the AC source was the same in Studies 1–3.

Study 1

A circuit was assembled using a $1,000\ \Omega$ (ohm) resistor, a 1 microfarad capacitor (*microfarad* [μF] is a unit of capacitance), 2 voltmeters for measuring output voltages, and an AC input voltage source (see Figure 1).



Key

- = wire (conducts current)
- ⓧ = voltmeter (measures voltage)
- ⚡ = resistor (impedes current flow)
- ⏏ = capacitor (stores charge)
- Ⓢ = AC input voltage source (voltage that varies regularly with time)

Figure 1

The students varied the frequency of the input voltage and measured the *output voltage*, the voltage across the resistor or the capacitor. At each frequency, they calculated the *gain*, the ratio of the measured output voltage to the measured input voltage. The results are shown in Table 1.

Table 1		
Frequency (Hz)	Gain across $1\ \mu\text{F}$ capacitor	Gain across $1,000\ \Omega$ resistor
10	1.00	0.06
10^2	0.85	0.53
10^3	0.16	0.99
10^4	0.02	1.00
10^5	0.00	1.00
10^6	0.00	1.00

Study 2

The students replaced the $1,000\ \Omega$ resistor with a $10,000\ \Omega$ resistor and repeated the procedure used in Study 1. The results are shown in Table 2.

Table 2		
Frequency (Hz)	Gain across $1\ \mu\text{F}$ capacitor	Gain across $10,000\ \Omega$ resistor
10	0.85	0.53
10^2	0.16	0.99
10^3	0.02	1.00
10^4	0.00	1.00
10^5	0.00	1.00
10^6	0.00	1.00

Study 3

The students used a $10\ \mu\text{F}$ capacitor and the $10,000\ \Omega$ resistor and repeated the procedure used in Study 1. The results are shown in Table 3.

Table 3		
Frequency (Hz)	Gain across $10\ \mu\text{F}$ capacitor	Gain across $10,000\ \Omega$ resistor
10	0.16	1.00
10^2	0.02	1.00
10^3	0.00	1.00
10^4	0.00	1.00
10^5	0.00	1.00
10^6	0.00	1.00

13. In Study 1, the same gain was obtained across the resistor and the capacitor at which, if any, of the frequencies tested?

A. 10^2 Hz
 B. 10^3 Hz
 C. 10^4 Hz
 D. None of the frequencies

14. Based on Studies 1 and 2 at 10^2 Hz, replacing the $1,000\ \Omega$ resistor with the $10,000\ \Omega$ resistor changed the measured output voltage across:

F. the resistor only.
 G. the capacitor only.
 H. both the resistor and the capacitor.
 J. neither the resistor nor the capacitor.



15. In Study 3, compared to the gain across the capacitor, the gain across the resistor was:
- A. always greater.
 - B. always smaller.
 - C. greater at high frequencies and less at low frequencies.
 - D. less at high frequencies and greater at low frequencies.
16. An *amplifier* produces a measured output voltage that is greater than the measured input voltage. Did the RC circuits used in the 3 studies amplify the input voltage at any of the frequencies tested?
- F. Yes, because all of the circuits produced a gain that was greater than 1.0 at one or more frequencies.
 - G. Yes, because all of the circuits produced a gain that was less than 1.0 at one or more frequencies.
 - H. No, because none of the circuits produced a gain that was greater than 1.0 at any frequency.
 - J. No, because none of the circuits produced a gain that was less than 1.0 at any frequency.
17. In Study 2 at 10^6 Hz, if the measured input voltage was 1.00 V (volt), the measured output voltage across the resistor was:
- A. 1.00 V.
 - B. 0.53 V.
 - C. 0.16 V.
 - D. 0.00 V.
18. In Study 1 with the output voltage measured across the resistor, a gain of 0.05 would most likely have been obtained at a frequency between:
- F. 1 Hz and 10 Hz.
 - G. 10 Hz and 10^2 Hz.
 - H. 10^2 Hz and 10^3 Hz.
 - J. 10^3 Hz and 10^4 Hz.

Passage IV

Burning household waste in metal barrels can release harmful *emissions* (airborne gases and particulates), including SO_2 , CO, NO_2 , dioxin, lead, and mercury. The remaining ash also may contain toxic substances. Two studies examined these emissions and ash.

Study 1

Two hundred pounds of household waste designated for burning was collected from 12 households. The waste was placed in a metal barrel located under a hood that collected emissions (see Figure 1). The waste was burned for 2 hours at 400°F to 500°F . The average reduction in the weight and the volume of the original waste after burning was 58% and 70%, respectively.

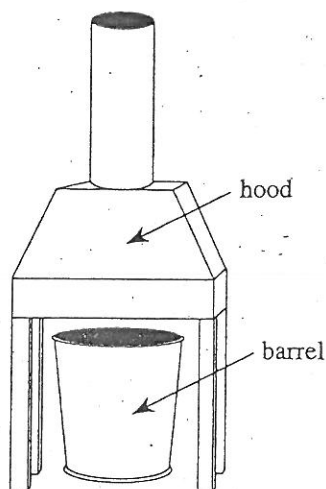


Figure 1

The concentrations of emissions, in parts per million (ppm), were measured. Table 1 shows the results and the established maximum allowed concentrations in emissions from *incinerators* (facilities that burn large quantities of household waste).

Table 1		
Emission	Measured concentration in barrel emissions (ppm)	Maximum allowed incinerator emission concentration (ppm)
I. Particulates	343,000	232,000
II. Gases		
SO_2	4.90	30
CO	149.20	100
NO_2	3.20	500
Dioxin	0.0000006	0.000025
Lead	0.004	0.41
Mercury	0.002	0.82

Study 2

The ash from the burned waste in Study 1 was analyzed for the presence of several substances, including mercury and lead. Table 2 shows the measured concentrations in the ash and the established limit that, if exceeded for any one of the substances, means the sample of ash would be considered hazardous.

Table 2		
Substance	Measured concentration (mg/L)	Established concentration limit (mg/L)
Arsenic	0.005	5.0
Barium	0.35	100
Cadmium	0.004	1.0
Chromium	0.04	5.0
Lead	0.01	5.0
Mercury	0.0004	0.2
Selenium	0.04	5.0

Figure and tables adapted from Chris Burger, "Emission Characteristics of Burn Barrels." ©1994 by Patrick Engineering, Inc.

19. Mercury is a liquid at room temperature. Based on the results for mercury in Study 1 and the design of that study, one would conclude which of the following about the properties of mercury? The temperatures of 400°F to 500°F that were reached in the metal barrel were sufficient to change some of the mercury from a:
 - A. liquid to a gas.
 - B. solid to a liquid.
 - C. liquid to a solid.
 - D. gas to a liquid.
20. The more toxic a substance, the smaller the concentration at which it is considered hazardous. According to this information and Study 1, which of the listed gases is considered the most toxic?
 - F. SO_2
 - G. CO
 - H. NO_2
 - J. Dioxin

21. If Study 2 had been repeated using the waste from 15 households instead of 12 households, the resulting measured concentration for each substance in the repeated Study 2 would most likely have been:
- A. much higher than the respective concentration in the original study.
 - B. similar to the respective concentration in the original study.
 - C. much lower than the respective concentration in the original study.
 - D. approximately zero for all substances.
22. According to the results of Study 1, which of the following gases emitted from burning waste in metal barrels exceeded the maximum allowed emission concentration set for incinerators?
- F. SO_2
 - G. CO
 - H. NO_2
 - J. Dioxin
23. Which of the following statements about the concentrations of substances in the ash from burning waste in metal barrels is supported by the results of Study 2? The ash left after burning waste in metal barrels:
- A. contained hazardous concentrations of all the tested substances.
 - B. contained hazardous concentrations of mercury and lead only.
 - C. did not contain hazardous concentrations of any of the tested substances.
 - D. did not contain measurable concentrations of mercury or lead.
24. It was hypothesized that burning waste in metal barrels in these studies would produce lower concentrations of particulates than does burning waste in incinerators that comply with established maximum concentrations. Is this hypothesis supported by the results of Study 1?
- F. Yes; the average concentration of particulates from burning waste in the metal barrels was lower than the maximum allowed particulate concentration from burning waste in incinerators.
 - G. Yes; the average concentration of particulates from burning waste in the metal barrels was higher than the maximum allowed particulate concentration from burning waste in incinerators.
 - H. No; the average concentration of particulates from burning waste in the metal barrels was lower than the maximum allowed particulate concentration from burning waste in incinerators.
 - J. No; the average concentration of particulates from burning waste in the metal barrels was higher than the maximum allowed particulate concentration from burning waste in incinerators.

Passage V

Albert Einstein predicted that light passing by a massive object would be deflected from its original direction of motion (Angle θ in Figure 1).

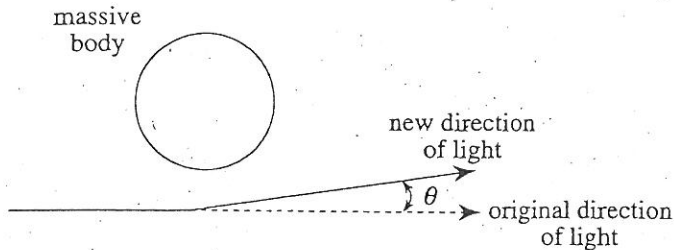


Figure 1

Einstein also predicted that light's travel time between certain planets would be increased by the Sun's gravity. Experiments 1 and 2 tested Einstein's predictions.

Experiment 1

Einstein predicted that for light of any wavelength skimming the visible edge of the Sun, $\theta = 1.75$ arcsec. To measure θ in visible light (wavelength = 0.00005 cm), scientists observed the stars appearing at the Sun's edge during a total solar eclipse, and they plotted the stars' positions in the sky. Next, they plotted the positions of the same stars when the stars did not appear close to the Sun; θ equaled the angle between the 2 positions of each star. An average of θ was found for all the stars tested. Then, scientists measured and found average values of θ for light at radio wavelengths (3.1 cm and 11.6 cm) skimming the Sun's visible edge.

The results are given in Table 1. (Note: $\pm$ signifies the uncertainty in a measurement. For example, at 0.00005 cm, the true value of θ was most likely between 1.57 and 1.87 arcsec.)

Table 1	
Wavelength (cm)	Average θ (arcsec)
0.00005	1.72 ± 0.15
3.1	1.77 ± 0.20
11.6	1.87 ± 0.30

Experiment 2

Scientists bounced radio waves off the surface of Venus as Venus moved around the Sun. Figure 2 shows the approximate positions of Venus (V) relative to the Sun (S) and Earth (E) on Days 0, 150, 300, and 450.

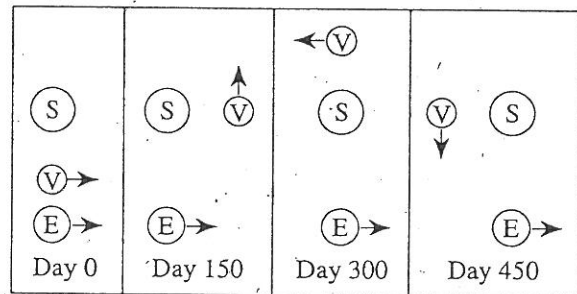


Figure 2

The delay measured by the scientists equaled the extra time the radio waves took to make the round trip between Earth and Venus because of the Sun's gravity (see Figure 3).

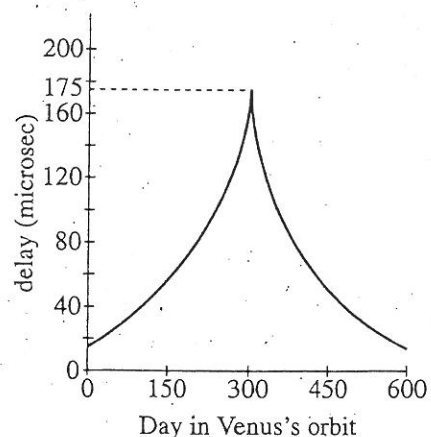
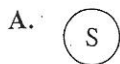


Figure 3

25. In Experiment 2 on Day 600, the relative positions of the Sun, Earth, and Venus were closest to those shown in which of the following diagrams?

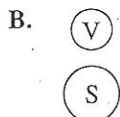


V

E



E



E



E

26. According to the uncertainty in the measurement of θ at a wavelength of 3.1 cm, the true value of θ was most likely between:

- F. 0.00 arcsec and 0.20 arcsec.
- G. 0.20 arcsec and 0.40 arcsec.
- H. 1.57 arcsec and 1.97 arcsec.
- J. 1.97 arcsec and 2.17 arcsec.

27. Suppose a spacecraft carrying a radio transmitter landed on Venus. Angle θ was measured on the same days as in Experiment 2 for the radio waves transmitted from the spacecraft to Earth. Based on Experiment 1, θ would most likely have been greatest on or about Day:

- A. 150.
- B. 300.
- C. 450.
- D. 600.

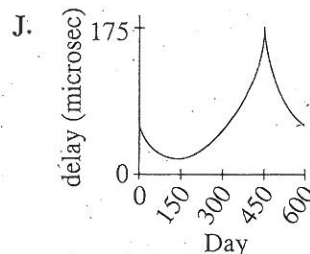
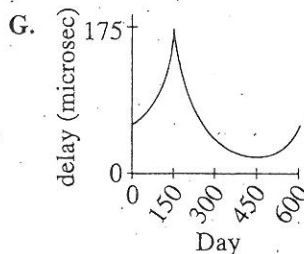
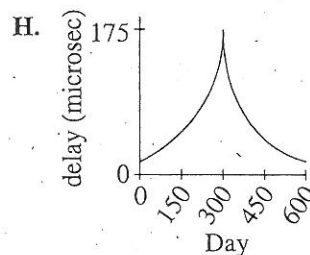
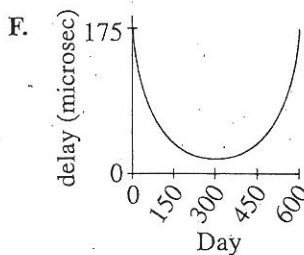
28. Suppose, on Day 0 in Experiment 2, the relative positions of the Sun, Earth, and Venus were as shown in the diagram below.

V

S

E

The graph of the delay versus Day would have looked most like which of the following graphs?



29. During Experiment 2, the largest delay in the radio waves was produced when the radio waves passed:

- A. farthest from the edge of the Moon both on their way to the Sun and on their return to Earth.
- B. closest to the edge of the Moon both on their way to the Sun and on their return to Earth.
- C. farthest from the edge of the Sun both on their way to Venus and on their return to Earth.
- D. closest to the edge of the Sun both on their way to Venus and on their return to Earth.

30. The Sun was used in both Experiments 1 and 2 most likely because the Sun:

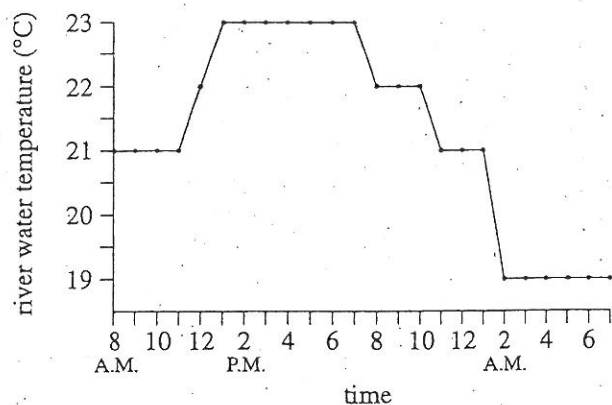
- F. is the brightest object in the solar system.
- G. emits both visible light and radio waves.
- H. is the most massive body in the solar system.
- J. is easily viewed during a solar eclipse.

Passage VI

The concentration of *dissolved oxygen* (DO) in river water affects the fish in the river. Many freshwater fish species, including crappie, perch, and bass, require a minimum average DO concentration of 5 parts per million (ppm) to thrive. Other fish, such as carp and gar, can tolerate average DO concentrations as low as 3 ppm. Table 1 shows the DO concentration at saturation for water at various temperatures. Figures 1 and 2 show the measured changes in a river's water temperature and DO concentration, respectively, over a 24-hour period in midsummer. Table 2 shows how the DO concentration affects fish species in a river.

Table 1	
Water temperature (°C)	DO concentration (ppm) at saturation*
0	14.6
5	12.8
10	11.3
15	10.2
20	9.2
25	8.4
30	7.7
35	7.0

*Saturation means the water contains the maximum amount of oxygen it can contain at that temperature under standard conditions.



Note: This river is located in a temperate region (cold winters, warm summers).

Figure 1

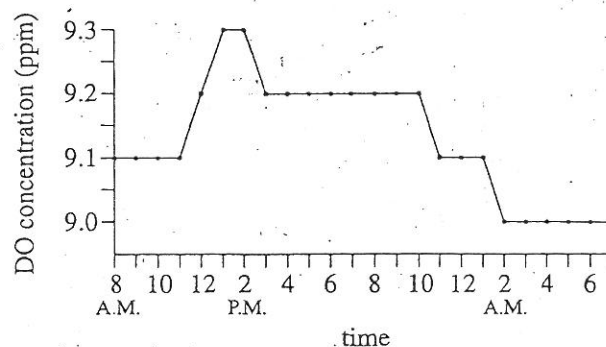


Figure 2

Table 2	
DO concentration (ppm)	Fish response
6–10	Fish of most species thrive.
4–5	Fish of many species stop feeding.
1–3	Fish of most species must come to the water surface for oxygen.
< 1	Fish die.

Table 1 and Figures 1 and 2 adapted from American Chemical Society, *ChemCom: Chemistry in the Community*, 2nd ed. ©1993 by American Chemical Society.

31. River water is often used in industrial processes and is then returned to the river much warmer than when it was removed. Based only on the information in Table 1, why would this warmed water be a potential danger to fish populations? The warmed river water most likely has:

- A. a higher DO concentration than does the cooler river water.
- B. a lower DO concentration than does the cooler river water.
- C. the same DO concentration as does the cooler river water.
- D. a DO concentration of zero.

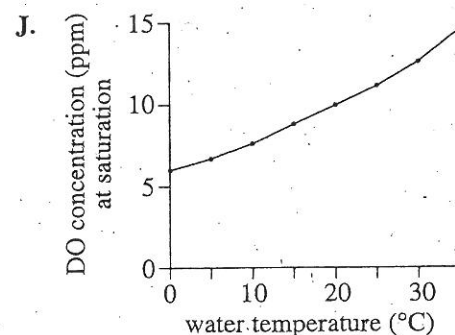
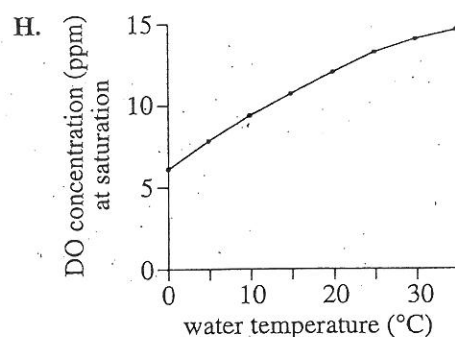
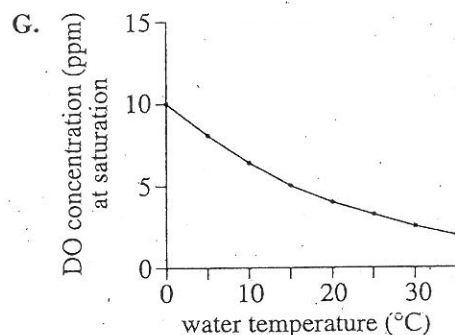
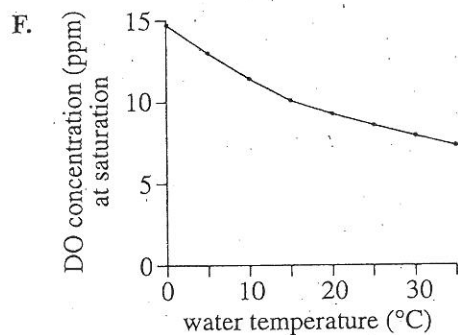
32. According to Figure 1 and Figure 2, which of the following best describes what happens to the water temperature and the DO concentration between 2 A.M. and 7 A.M.?

- | | Water temperature | DO concentration |
|----|--------------------|--------------------|
| F. | Steadily increases | Remains the same |
| G. | Steadily decreases | Steadily decreases |
| H. | Remains the same | Remains the same |
| J. | Remains the same | Steadily increases |

33. According to Table 1 and Table 2, river water at a temperature in which of the following ranges would most likely have a low enough DO concentration to endanger most fish species living in that water?

- A. Less than 25°C
- B. Between 25°C and 30°C
- C. Between 30°C and 35°C
- D. Greater than 35°C

34. Which of the following graphs best represents the relationship between water temperature and DO concentration at saturation shown in Table 1?



35. According to Figure 1 and Figure 2, what were the measured water temperature and DO concentration of the river water at 10 P.M.?

- A. 21°C and 9.3 ppm
- B. 22°C and 9.3 ppm
- C. 22°C and 9.2 ppm
- D. 23°C and 9.1 ppm

Passage VII

The *male-to-female body mass ratio* (MFMR) of a species can be calculated as follows:

$$\text{MFMR} = M_m \div F_m$$

In this equation, M_m is the average body mass of adult males and F_m is the average body mass of adult females.

The figure shows the MFMR of some species within 4 primate groups: apes and humans; Old World monkeys; New World monkeys; and prosimians. A species with an MFMR of 1 is *monomorphic* with respect to body mass. A species with an MFMR different than 1 is *dimorphic* with respect to body mass. (Note: In the figure, MFMR values are calculated to the nearest tenth.)

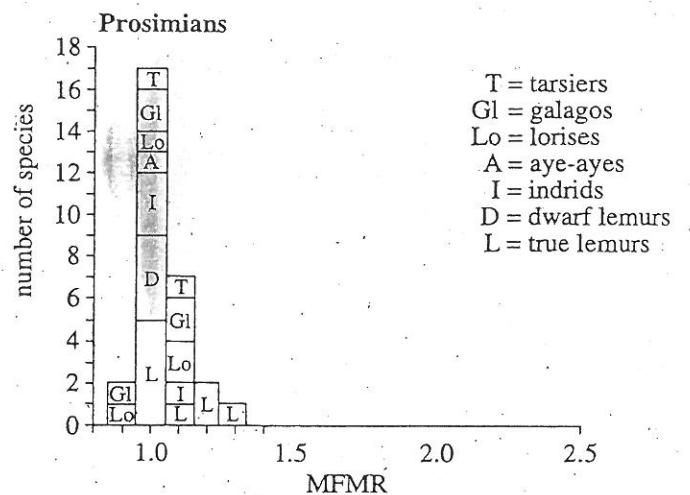
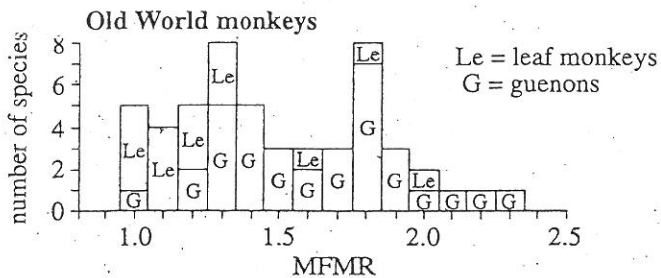
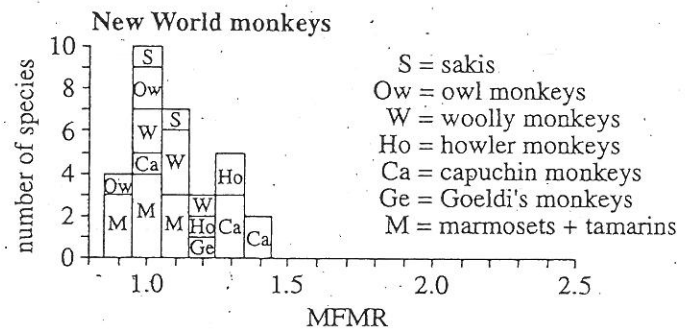
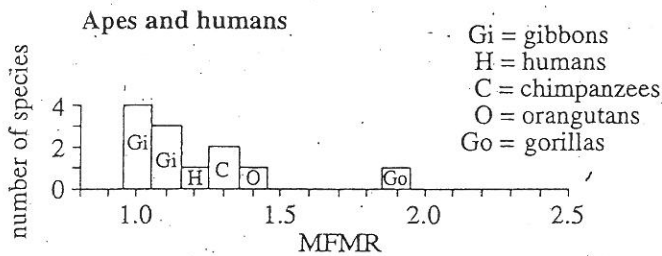


Figure adapted from Steve Jones, Robert Martin, and David Pilbeam, eds., *The Cambridge Encyclopedia of Human Evolution*. ©1992 by Cambridge University Press.



36. According to the figure, the body mass of the average adult male orangutan is how many times greater than the body mass of the average adult female orangutan?
- F. 1.1
G. 1.4
H. 1.6
J. 1.9
37. Which of the following statements about the relative size of male and female chimpanzees and aye-ayes is consistent with the information provided?
- A. Both chimpanzees and aye-ayes are dimorphic with respect to body mass.
B. Both chimpanzees and aye-ayes are monomorphic with respect to body mass.
C. Chimpanzees are dimorphic with respect to body mass, and aye-ayes are monomorphic with respect to body mass.
D. Chimpanzees are monomorphic with respect to body mass, and aye-ayes are dimorphic with respect to body mass.
38. According to the figure, which of the following primate groups has the largest proportion of monomorphic species?
- F. Apes
G. Old World monkeys
H. New World monkeys
J. Prosimians
39. On average, which of the following types of apes shown in the figure has the greatest MFMR ?
- A. Gibbons
B. Chimpanzees
C. Orangutans
D. Gorillas
40. Which of the following best explains the units of measure that are associated with the MFMR ?
- F. The numerator is in units of mass and the denominator is in units of mass, so the ratio has no units.
G. The numerator is in units of mass and the denominator is in units of mass, so the ratio is in units of mass.
H. The numerator is in units of volume and the denominator is in units of volume, so the ratio has no units.
J. The numerator is in units of volume and the denominator is in units of volume, so the ratio is in units of volume.

END OF TEST 4

STOP! DO NOT RETURN TO ANY OTHER TEST.

Form 10G
ACT Assessment Writing Test Prompt
(June 2005)

Educators often discuss whether high school sports have a positive influence on students. Some educators think high school sports do have a positive influence because the lessons learned from athletic competition add to the lessons learned in the classroom. Other educators think high school sports do not have a positive influence because the emphasis on sports often overshadows student achievement in other areas. In your opinion, do high school sports have a positive influence on students?

In your essay, take a position on this question. You may write about either one of the two points of view given, or you may present a different point of view on this question. Use specific reasons and examples to support your position.

Form 61D (June 2005)
Finding Tomatoes in Winter

English

- | | |
|-------|-------|
| 1) D | 51) B |
| 2) G | 52) J |
| 3) C | 53) C |
| 4) J | 54) G |
| 5) C | 55) A |
| 6) J | 56) H |
| 7) A | 57) D |
| 8) F | 58) J |
| 9) C | 59) A |
| 10) H | 60) H |
| 11) B | 61) D |
| 12) F | 62) H |
| 13) B | 63) B |
| 14) F | 64) G |
| 15) C | 65) C |
| 16) H | 66) G |
| 17) D | 67) A |
| 18) J | 68) J |
| 19) A | 69) A |
| 20) G | 70) G |
| 21) C | 71) A |
| 22) J | 72) J |
| 23) C | 73) B |
| 24) F | 74) J |
| 25) A | 75) C |
| 26) G | |
| 27) C | |
| 28) H | |
| 29) D | |
| 30) J | |
| 31) C | |
| 32) J | |
| 33) B | |
| 34) H | |
| 35) A | |
| 36) J | |
| 37) D | |
| 38) G | |
| 39) B | |
| 40) F | |
| 41) D | |
| 42) G | |
| 43) D | |
| 44) F | |
| 45) A | |
| 46) F | |
| 47) C | |
| 48) J | |
| 49) A | |
| 50) G | |

Math

- | | |
|-------|-------|
| 1) A | 51) A |
| 2) H | 52) J |
| 3) C | 53) E |
| 4) J | 54) G |
| 5) E | 55) E |
| 6) H | 56) G |
| 7) B | 57) E |
| 8) G | 58) G |
| 9) D | 59) A |
| 10) K | 60) J |
| 11) C | |
| 12) H | |
| 13) D | |
| 14) K | |
| 15) D | |
| 16) H | |
| 17) E | |
| 18) J | |
| 19) C | |
| 20) J | |
| 21) C | |
| 22) G | |
| 23) A | |
| 24) J | |
| 25) A | |
| 26) F | |
| 27) D | |
| 28) H | |
| 29) C | |
| 30) K | |
| 31) E | |
| 32) K | |
| 33) B | |
| 34) H | |
| 35) B | |
| 36) G | |
| 37) D | |
| 38) K | |
| 39) A | |
| 40) J | |
| 41) B | |
| 42) H | |
| 43) E | |
| 44) J | |
| 45) A | |
| 46) G | |
| 47) A | |
| 48) H | |
| 49) A | |
| 50) F | |

Reading

- | |
|-------|
| 1) C |
| 2) F |
| 3) C |
| 4) J |
| 5) A |
| 6) F |
| 7) B |
| 8) G |
| 9) D |
| 10) H |
| 11) D |
| 12) H |
| 13) B |
| 14) G |
| 15) C |
| 16) J |
| 17) D |
| 18) J |
| 19) C |
| 20) F |
| 21) C |
| 22) H |
| 23) C |
| 24) J |
| 25) A |
| 26) F |
| 27) B |
| 28) J |
| 29) C |
| 30) J |
| 31) A |
| 32) J |
| 33) B |
| 34) J |
| 35) B |
| 36) H |
| 37) A |
| 38) H |
| 39) D |
| 40) G |

Science

- | |
|-------|
| 1) B |
| 2) H |
| 3) D |
| 4) G |
| 5) D |
| 6) G |
| 7) D |
| 8) G |
| 9) C |
| 10) F |
| 11) D |
| 12) H |
| 13) D |
| 14) H |
| 15) A |
| 16) H |
| 17) A |
| 18) F |
| 19) A |
| 20) J |
| 21) B |
| 22) G |
| 23) C |
| 24) J |
| 25) A |
| 26) H |
| 27) B |
| 28) F |
| 29) D |
| 30) H |
| 31) B |
| 32) H |
| 33) D |
| 34) F |
| 35) C |
| 36) G |
| 37) C |
| 38) J |
| 39) D |
| 40) F |

ACT Resource Links

ACT Online Practice Tests: <https://www.crackab.com/act/>

✧ **ACT English Practice Tests:**

<https://www.crackab.com/act/english/>

✧ **ACT Math Practice Tests:**

<https://www.crackab.com/act/math/>

✧ **ACT Reading Practice Tests:**

<https://www.crackab.com/act/reading/>

✧ **ACT Science Practice Tests:**

<https://www.crackab.com/act/science/>

ACT Grammar: <https://www.crackab.com/act/grammar/>

ACT Real Past Papers Download:

<https://www.crackab.com/act-downloads/>

Digital SAT & New SAT Practice Tests:

<https://www.cracksat.net>

Real SAT Tests Download:

<http://www.cracksat.net/sat-downloads/>

AP Exams Practice Tests:

<https://www.crackap.com>

<https://www.apstudy.net>

Explanation of Procedures Used to Obtain Scale Scores from Raw Scores

On each of the four tests on which you marked any responses, the total number of correct responses yields a raw score. Use the table below to convert your raw scores to scale scores. For each test, locate and circle your raw score or the range of raw scores that includes it in the table below. Then, read across to either outside column of the table and circle the scale score that corresponds to that raw score. As you determine your scale scores, enter them in the blanks provided on the right. The highest possible scale score for each test is 36. The lowest possible scale score for any test on which you marked any responses is 1.

Next, compute the Composite score by averaging the four scale scores. To do this, add your four scale scores and divide the sum by 4. If the resulting number ends in a fraction, round it off to the nearest whole number. (Round down any fraction less than one-half; round up any fraction that is one-half or more.) Enter this number in the blank. This is your Composite score. The highest possible Composite score is 36. The lowest possible Composite score is 1.

ACT Test 61D Your Scale Score

English _____

Mathematics _____

Reading _____

Science _____

Sum of scores _____

Composite score (sum ÷ 4) _____

NOTE: If you left a test completely blank and marked no items, do not list a scale score for that test. If any test was completely blank, do not calculate a Composite score.

Scale Score	Raw Scores				Scale Score
	Test 1 English	Test 2 Mathematics	Test 3 Reading	Test 4 Science	
36	75	60	40	40	36
35	73-74	58-59	39	39	35
34	72	56-57	38	--	34
33	71	55	37	38	33
32	70	54	--	37	32
31	69	53	36	--	31
30	68	51-52	35	36	30
29	66-67	49-50	34	35	29
28	64-65	47-48	33	34	28
27	63	44-46	31-32	33	27
26	61-62	42-43	30	31-32	26
25	58-60	39-41	29	29-30	25
24	56-57	37-38	27-28	28	24
23	54-55	35-36	26	26-27	23
22	51-53	34	25	24-25	22
21	48-50	32-33	23-24	22-23	21
20	45-47	30-31	22	20-21	20
19	42-44	28-29	20-21	18-19	19
18	39-41	25-27	19	16-17	18
17	37-38	22-24	18	15	17
16	34-36	18-21	16-17	14	16
15	31-33	15-17	15	12-13	15
14	28-30	12-14	13-14	11	14
13	26-27	10-11	12	10	13
12	25	08-09	10-11	09	12
11	23-24	06-07	08-09	08	11
10	21-22	05	07	07	10
9	19-20	04	06	05-06	9
8	16-18	--	05	04	8
7	13-15	03	04	--	7
6	11-12	--	--	03	6
5	08-10	02	03	02	5
4	06-07	--	02	--	4
3	04-05	01	--	01	3
2	03	--	01	--	2
1	00-02	00	00	00	1